Agentic Image Processing: A Systematic Review of Closed-Loop Image Transformation Systems

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ABSTRACT

Image restoration, enhancement, editing, and controllable generation are increasingly performed by systems that do more than run a single network. In these systems, a controller diagnoses the image, decomposes the request, selects and configures tools, inspects intermediate results, and revises the result or stops. The literature is spread across several communities, and the word *agent* is applied to systems with very different degrees of runtime control. This review defines agentic image processing by its behavior: an identifiable controller uses observations of the input or of an intermediate image to choose later processing actions. We systematically search the literature and analyze representative, high-impact systems published since 2017 in terms of their tasks, controllers, observations, actions, feedback, memory, organization, training, and degree of autonomy. We propose a four-axis taxonomy and a five-level autonomy scale (A0–A4), describe a common perceive–plan–act–verify–remember architecture, and compare restoration, retouching, editing, closed-loop generation, workflow composition, and domain applications. Three quarters of these systems appeared in 2025 or 2026, a growth driven by the development of multimodal large language models that can perceive images, plan, and call tools. Controllers are moving from training-free orchestration toward reinforcement learning, and nearly all systems rely on learned or language-model judges whose errors propagate through the loop. We introduce a six-layer evaluation framework that separates image quality, goal satisfaction, decision quality, trajectory quality, efficiency, and human-centered reliability, together with a reporting checklist. We close with design guidelines and open problems in evaluator robustness, credit assignment, tool interoperability, memory, efficiency, provenance, and safety.

1. Introduction

For most of the deep-learning era, progress in image processing has been measured by better mappings. A denoiser, a deraining network, a super-resolution model, or an instruction-following editor receives an image and returns another image in a single forward computation (Zhang et al., 2017; Zamir et al., 2022; Liang et al., 2021; Brooks et al., 2023). All-in-one restoration models and large generative editors have broadened what one model can handle (Li et al., 2022; Potlapalli et al., 2023; Conde et al., 2024; Black Forest Labs et al., 2025; Wu et al., 2025), yet a practical request rarely reduces to one mapping. A photograph may contain haze, sensor noise, and compression artifacts whose removal order matters. A retouching request may be aesthetic and underspecified. A multi-step edit must change some content while preserving everything else, and a professional workflow may require tools that no single model provides. In each case someone must decide *what* is wrong, *which* operation to apply, *in what order*, *with which parameters*, and *when to stop*. Traditionally, that someone was a human expert.

A rapidly growing class of systems moves these decisions into software. RestoreAgent fine-tunes a multimodal large language model (MLLM) to emit an ordered program of restoration models and to decide when to roll back or terminate (Chen et al., 2024). AgenticIR combines a vision–language perceiver with a language-model scheduler that executes one tool at a time, reflects on the intermediate result, rolls back failures, and reschedules (Zhu et al., 2025b). JarvisArt and JarvisEvo translate retouching intent into editable Adobe Lightroom operations and train the controller with reinforcement learning (Lin et al., 2025b,c). GenArtist organizes generation and editing tools into a verification tree (Wang et al., 2024c), and CoSTA* couples task decomposition by a language model with A* search over a graph of tools (Gupta et al., 2025). Other agents now assemble complete workflows for ComfyUI, a node-based interface for diffusion models (Xue et al., 2025; Li et al., 2026b). The object being designed in these papers is no longer only a network but a *decision process over image states*, and a recent position paper argues that such systems should become a central research direction for image processing (Gu, 2025).

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Table 1
Positioning relative to existing reviews. ✓ = central, ∘ = partial or illustrative, – = absent. Scope abbreviations: IR restoration, EN enhancement and retouching, ED editing, GN closed-loop generation, WF workflow composition.

Review	Type	IR	EN	ED	GN	WF	Main organizing principle
Ulhaq (2025)	Review	∘	–	∘	∘	–	Autonomy across all computer vision, including perception and navigation
Gu (2025)	Position	✓	∘	∘	–	–	Model-centric versus system-centric image processing, with capability levels
Sun et al. (2026b)	Survey	✓	–	–	–	–	Evolution from specialized to agentic restoration
Yang et al. (2026)	Survey	✓	–	–	–	–	Controller families and protocol-aware restoration evaluation
Xing et al. (2026)	Survey	∘	∘	✓	✓	∘	Agentic generation across image, video, 3D, documents, and user interfaces
Mei et al. (2026)	Survey	–	∘	✓	✓	✓	Production roles in creative visual content
Jiang et al. (2025) and Huang et al. (2025c)	Surveys	✓	–	✓	–	–	Model-centric all-in-one restoration and diffusion-based editing
This review	Systematic review	✓	✓	✓	✓	✓	Behavioral definition, four-axis taxonomy, design patterns shared across tasks, and process-aware evaluation

This shift matters for research on expert and intelligent systems. Agentic image processing (AIP) systems combine knowledge representation (descriptions of tools, priors about degradations, stored experience), search and planning (greedy selection, depth-first search with rollback, A* search, Monte Carlo tree search), learned perception (quality assessment and vision–language diagnosis), and sequential decision making under uncertainty. They therefore inherit classical questions about autonomy, verification, and credit assignment (Wooldridge and Jennings, 1995; Franklin and Graesser, 1997; Russell and Norvig, 2021) in a domain where outcomes are images and correctness is often perceptual. They also bring new failure modes. A controller that trusts a miscalibrated judge can optimize the judge instead of the image, and a long sequence of operations can accumulate artifacts that a single model would never produce (Pan et al., 2024; Tang et al., 2026).

The literature is nevertheless difficult to navigate. First, it is fragmented. Restoration agents are published in low-level vision venues, retouching agents in graphics and multimedia, editing and generation agents in machine learning and language venues, and workflow agents in system demonstrations. Second, terminology is unreliable. "Agent" may denote an autonomous controller, a role prompt in a fixed pipeline, a reinforcement-learning policy, or simply a module. Conversely, earlier reinforcement-learning systems such as RL-Restore, PixelRL, and Exposure implement genuine loops of states, actions, and feedback without modern agent vocabulary (Yu et al., 2018; Furuta et al., 2020; Hu et al., 2018). Third, evaluation remains almost entirely focused on the output image. Most papers report the peak signal-to-noise ratio (PSNR), no-reference quality scores, or the verdicts of vision–language judges on the final image, even though the claimed contribution is a better decision process.

Existing reviews cover parts of this space (Table 1). Broad surveys of agentic computer vision treat image transformation as one outcome among perception, navigation, and interface control (Ulhaq, 2025). Surveys of agentic visual generation and creation emphasize content production across images, video, 3D, and documents (Xing et al., 2026; Mei et al., 2026). Restoration-specific reviews trace the transition from specialized to agentic restorers and analyze controller families and evaluation protocols, but stop at restoration (Sun et al., 2026b; Yang et al., 2026). Model-centric surveys of all-in-one restoration and diffusion-based editing are indispensable for understanding the tools that agents call, but they do not analyze runtime control (Jiang et al., 2025; Huang et al., 2025c; Shuai et al., 2024). What is missing is a review that treats restoration, enhancement, editing, and closed-loop generation as instances of *one* decision problem, describes the systems with a common vocabulary, and asks how their agentic claims should be evaluated.

This review addresses five research questions:

RQ1: What observable behavior distinguishes an image-processing agent from a fixed pipeline, a prompt-conditioned model, or generic multimodal reasoning?

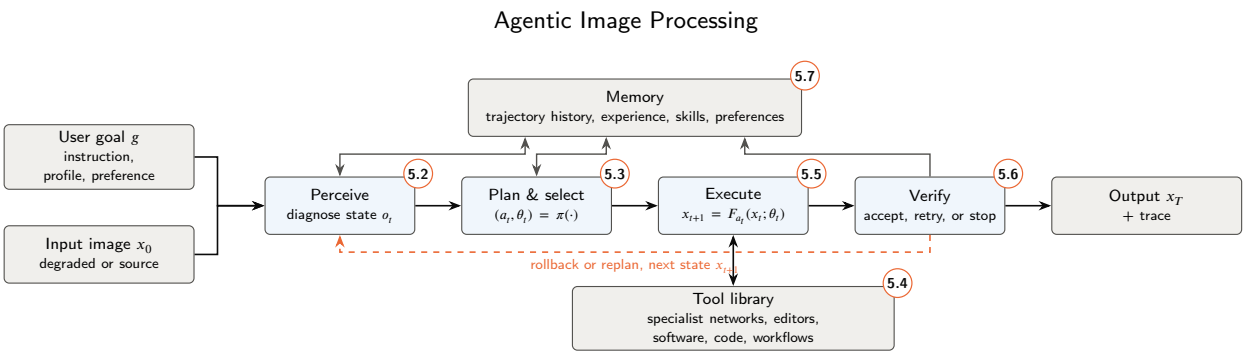


Figure 1: The agentic image-processing loop used throughout this review. A controller receives a goal and observations of the current image state x_t, selects an action a_t with parameters θ_t from a tool library, executes it, and verifies the outcome. Verification can lead to acceptance, retry, rollback, replanning, or stopping. Memory keeps information within an episode or across tasks. Numbers in circles indicate the sections in which each component is analyzed.

RQ2: How do current systems represent image state, plan transformations, select and configure tools, verify outcomes, and accumulate experience?

RQ3: How do these mechanisms differ across restoration, enhancement, editing, closed-loop generation, workflow composition, and application domains?

RQ4: Which benchmarks and metrics can support claims about decisions and trajectories rather than only about output images?

RQ5: Which technical, methodological, and deployment issues limit reliable, efficient, and reproducible agentic image processing?

Our contributions are as follows. (i) We give an operational definition of AIP based on observable behavior, together with an autonomy scale from A0 to A4 that does not depend on the number of named agents. (ii) We systematically search the literature and analyze representative systems across five task families with a common set of descriptive dimensions. (iii) We describe a common perceive–plan–act–verify–remember architecture and analyze the design alternatives for each of its components, including how controllers are trained. (iv) We compare these mechanisms across task families and application domains, drawing on the quantitative evidence reported in the original papers. (v) We propose a six-layer evaluation framework, a map from metrics to the claims they can support, and a reporting checklist, and we derive design guidelines and a research agenda.

The rest of the paper is organized around the decision loop shown in Fig. 1. Section 2 traces the research lines from which agentic image processing emerged, and Section 3 defines the concept and the scope of the review. Section 4 presents the taxonomy and the autonomy scale. Section 5 analyzes each component of the loop, Sections 6 and 7 compare tasks and application domains, and Section 8 discusses evaluation. Sections 9 and 10 derive design guidelines and open problems, and Section 11 concludes.

2. Background: four lineages that converge in agentic image processing

Agentic image processing did not arise from a single idea. It is the meeting point of four research lines: capable image-to-image models that serve as tools, sequential decision making over image operations, language agents that plan and call tools, and models that can judge image quality and instruction compliance. Figure 2 places representative milestones on a common time axis.

2.1. Model-centric image processing: the tools

Deep restoration networks progressed from residual convolutional denoisers such as DnCNN (Zhang et al., 2017) to transformer restorers such as SwinIR and Restormer (Liang et al., 2021; Zamir et al., 2022). They also moved from regression to generative priors for realistic super-resolution and blind restoration, as in Real-ESRGAN, GFPGAN, SR3, DDRM, DiffBIR, and SUPIR (Wang et al., 2021b,a; Saharia et al., 2022; Kawar et al., 2022; Lin et al., 2024;

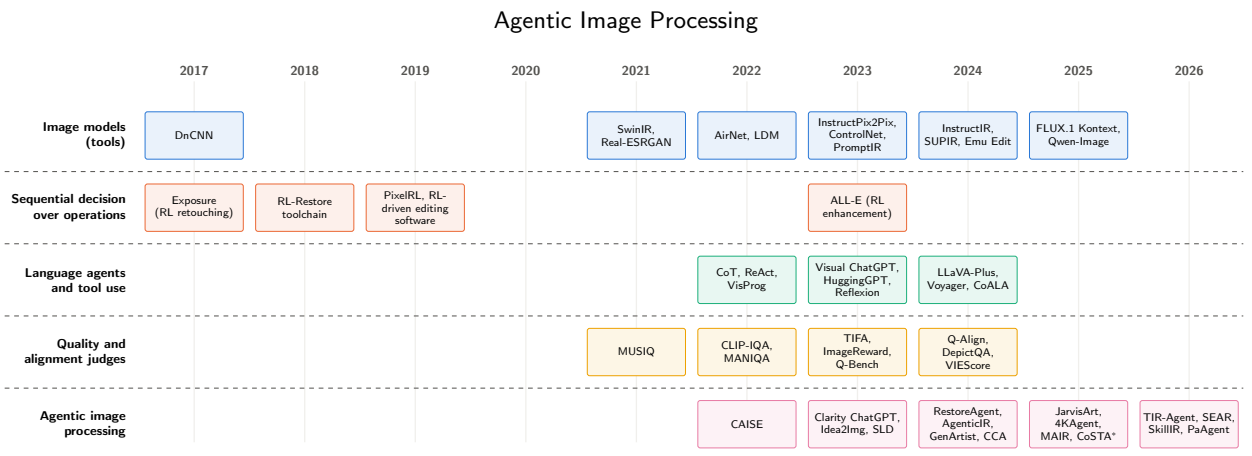


Figure 2: Milestones of the four lineages (rows) that converge in agentic image processing (bottom row). Positions are by year of first public release.

Yu et al., 2024). All-in-one models such as AirNet and PromptIR then handled several degradations with one network by learning degradation representations or prompts (Li et al., 2022; Potlapalli et al., 2023), and InstructIR conditioned restoration on natural-language instructions (Conde et al., 2024). In parallel, editing moved from attention manipulation in Prompt-to-Prompt (Hertz et al., 2023) to instruction-tuned editors such as InstructPix2Pix and Emu Edit (Brooks et al., 2023; Sheynin et al., 2024), to editors guided by MLLMs (Fu et al., 2024; Huang et al., 2024), and finally to large general-purpose editing and unified models such as Step1X-Edit, FLUX.1 Kontext, Qwen-Image, and BAGEL (Liu et al., 2025c; Black Forest Labs et al., 2025; Wu et al., 2025; Deng et al., 2025). Generation became controllable through latent diffusion models (LDM), SDXL, and spatial conditioning with ControlNet (Rombach et al., 2022; Podell et al., 2024; Zhang et al., 2023b).

These models are the actions of an agent, and their limitations explain why control is needed. Each operates in a fixed inference regime. All-in-one models trade per-task performance for coverage and degrade on combinations of degradations not seen in training (Jiang et al., 2025). The order of operations on mixed degradations matters (Zhu et al., 2025b; Jiang et al., 2026b), and restoration faces a trade-off between perceptual quality and distortion that no single operating point resolves for every user (Blau and Michaeli, 2018). Models conditioned on prompts or degradation embeddings adapt internally, but they expose no decision that can be inspected, revised, or reused. In the taxonomy of Section 4 they are therefore tools (level A0) rather than agents.

2.2. Sequential decision making over image operations

The idea that image processing is a sequence of decisions predates language models. RL-Restore trained a deep reinforcement-learning (RL) agent to select, at each step, one tool from a toolbox of small restoration networks, including a stopping action, so that mixed distortions are removed by a learned, input-dependent chain of tools (Yu et al., 2018). PixelRL assigned an agent to every pixel and let a fully convolutional policy choose per-pixel filters over several steps for denoising, restoration, and local color enhancement (Furuta et al., 2019, 2020). For photo retouching, Exposure modelled post-processing as a sequence of differentiable, resolution-independent filters chosen by a policy trained with reinforcement learning and adversarial learning from unpaired data (Hu et al., 2018), and Kosugi and Yamasaki (2020) trained a policy to operate conventional editing software. Later work used reinforcement learning for low-light enhancement, guided by aesthetics in ALL-E (Li et al., 2023) and by an asynchronous actor–critic (A3C) agent in Mahdi (2024), and for task-driven enhancement before medical segmentation (Wang et al., 2025b).

This lineage contributes the formal vocabulary of states, actions, rewards, and termination, together with actor–critic and policy-gradient training methods such as A3C and proximal policy optimization (PPO) (Mnih et al., 2016; Schulman et al., 2017). Its limitations are equally instructive. Action spaces were small and fixed at training time, observations were learned features without semantics, and rewards required paired data or hand-crafted proxies. Language-model controllers relaxed the first two constraints, and recent MLLM agents trained with reinforcement learning (Section 5.8) bring the third back to the fore.

2.3. Language agents, tool use, and reflection

Large language models (LLMs) acquired the ability to reason step by step through chain-of-thought (CoT) prompting and self-consistency (Wei et al., 2022; Wang et al., 2023b). ReAct interleaved reasoning with actions (Yao et al., 2023b), Toolformer learned when to call tools (Schick et al., 2023), and Self-Refine and Reflexion improved outputs through self-feedback and verbal reflection (Madaan et al., 2023; Shinn et al., 2023). Tree of Thoughts added search over candidate reasoning steps (Yao et al., 2023a), and Voyager maintained a skill library for lifelong improvement (Wang et al., 2024a). The CoALA framework organizes these capabilities into memory, action space, and decision procedure (Sumers et al., 2024), and several surveys review language and multimodal agents in general (Wang et al., 2024b; Xi et al., 2025; Xie et al., 2025; Guo et al., 2024).

Vision entered through tool orchestration. Visual ChatGPT connected a language model to visual foundation models through a prompt manager, enabling conversational drawing and editing (Wu et al., 2023a). HuggingGPT planned over models hosted on a public model hub (Shen et al., 2023), MM-REACT prompted a language model to invoke vision experts (Yang et al., 2023), and VisProg and ViperGPT generated programs that compose vision modules (Gupta and Kembhavi, 2023; Surís et al., 2023). Visual instruction tuning produced MLLMs such as LLaVA and Qwen2.5-VL that perceive images directly (Liu et al., 2023; Bai et al., 2025), and LLaVA-Plus trained them to use tools (Liu et al., 2024a). Most of these early systems targeted understanding. Image transformation was one tool family among many, and the output image was rarely inspected before being returned. Image-processing agents specialize this machinery to transitions between image states, where the critical question is not whether an answer is correct but whether the image improved and what was preserved.

2.4. Machines that judge images

A loop can only be closed if the system can evaluate its own intermediate results. Classical no-reference image quality assessment (NR-IQA) models such as MUSIQ, CLIP-IQA, and MANIQA (Ke et al., 2021; Wang et al., 2023a; Yang et al., 2022), and learned models of human preference for generated images such as ImageReward and HPSv2 (Xu et al., 2023; Wu et al., 2023b), supply scalar scores. MLLMs have been benchmarked and tuned for low-level perception in Q-Bench and Q-Instruct (Wu et al., 2024a,b), for score regression in Q-Align (Wu et al., 2024c), and for descriptive and comparative quality reasoning in DepictQA (You et al., 2024, 2025). Question-answering metrics such as TIFA and MLLM-based judges such as VIEScore assess the faithfulness of generated images to text and the quality of edits (Hu et al., 2023; Ku et al., 2024a). AgenticIQA turns quality assessment itself into a planned, tool-using process (Zhu et al., 2025a), and the broader practice of using an LLM as a judge supplies pairwise and rubric-based verdicts (Zheng et al., 2023). Almost every closed-loop system discussed in this review uses one or more of these judges for verification, candidate selection, or reward. As Section 8 shows, the reliability of agentic image processing is consequently bounded by the reliability of its judges.

2.5. Convergence

From 2023 onward these lineages merged. Language and multimodal models became controllers, restoration, editing, and generation models became tools, and quality and alignment judges became verifiers and rewards. Reinforcement learning returned as a way to train the controller itself. Early representatives were the conversational editor CAISE (Kim et al., 2022), Clarity ChatGPT for restoration and enhancement (Wei et al., 2023), Idea2Img and self-correcting LLM-controlled diffusion (SLD) for generation (Yang et al., 2024b; Wu et al., 2024d), and CCA, GenArtist, RestoreAgent, and AgenticIR for editing and restoration (Hang et al., 2025; Wang et al., 2024c; Chen et al., 2024; Zhu et al., 2025b). The rapid growth that followed coincides with the arrival of capable open and proprietary MLLMs, which made it possible for a single model to look at an image, reason about it, and call tools. The remainder of this review analyzes the systems that followed.

3. Definition and scope of the review

3.1. Operational definition

Let x_t denote the image state at step t (with x_0 the input), g the goal (an instruction, a quality profile, or an implicit objective such as "restore"), h_t the interaction history, and m an optional memory that may persist across episodes. A controller forms an observation

$$o_t = P(x_t, h_t, g), \tag{1}$$

selects an action and its parameters

$$(a_t, \theta_t) = \pi\left(g, o_{\le t}, m\right), \qquad a_t \in \mathcal{A} \cup \{\text{STOP}, \text{ROLLBACK}\}, \tag{2}$$

and an executor produces the next state

$$x_{t+1} = F_{a_t}(x_t; \theta_t). \tag{3}$$

The action set $\mathcal{A}$ may contain specialist networks, generative editors, operations of professional software, code, or edits of a workflow graph. The parameters θ_t may include the choice of model, the region, the strength, the prompt, or numeric settings. Equations (1)–(3) describe a partially observed sequential decision problem (Sutton and Barto, 2018). The controller rarely sees the true degradation or the user's full intent, and the reward (perceptual quality, instruction satisfaction, preservation of content) is usually available only through a learned or human judge.

> **Definition.** A system performs *agentic image processing* if an identifiable controller uses an observation of the input or an intermediate image to determine a later image-processing action, tool, parameter, region, candidate selection, retry, rollback, or stopping decision, and the controlled outcome is an image (or an editable image representation).

The definition has three consequences. First, the outcome must be an image transformation. Systems that answer questions, write captions, retrieve images, or navigate are excluded even if they reason about images. Second, there must be a non-trivial set of possible actions with a meaningful counterfactual, that is, a different observation could have led to a different operation, order, parameter, or stop. Third, agency is established by system behavior, not by vocabulary. A reinforcement-learning restorer that chooses tools step by step is included, whereas a fixed chain of prompts labelled "multi-agent" is classified by what it actually decides. We call a system *closed-loop* when an observation made *after* an executed action can change a subsequent action. This property is the core of autonomy levels A3 and A4 (Section 4.3).

3.2. Literature collection

We searched for papers that combine agent-related terms (agent, agentic, multi-agent, tool use, planning, reflection) with image-processing terms (restoration, enhancement, super-resolution, retouching, editing, inpainting, generation), and complemented these searches with domain-specific ones in medical imaging, remote sensing, autonomous driving, and photography. We also examined recent proceedings of the main conferences in computer vision, machine learning, multimedia, and artificial intelligence, and followed the references of related surveys and of the included systems, both backward and forward. Because the field changes quickly, the search was repeated shortly before submission.

From the candidates we selected representative and influential systems, namely those that introduced a new form of runtime control, that are widely cited or used as baselines, or that extend agentic processing to a new task or application domain. The selection is not meant to be exhaustive. Its purpose is to cover the main design patterns, and the counts reported below describe this selection rather than the whole field. Tools, benchmarks, and foundational works are cited where they help to explain the selected systems.

3.3. Inclusion criteria

A system was included when its output is an image or an editable image representation, such as a layered design or a workflow that produces an image. It had to contain an identifiable controller, which may be a learned policy, a language or multimodal model, a search procedure, or a team of agents with distinct roles, and this controller had to make runtime decisions among several possible operations. The paper also had to describe these decisions in enough detail for us to determine what the controller observes, which actions it can take, and what feedback it receives.

We excluded systems whose agency concerns only perception or text output, such as agents for visual question answering, captioning, retrieval, graphical user interfaces, or embodied navigation. We also excluded works in which "agent" names a component of a network, and systems in which a language model rewrites a prompt once without seeing any image, unless it chooses among tools or models on the basis of what it observes. Systems that choose a model or a parameter setting once from the input and never revise it (level A1 in Section 4.3) were kept as boundary cases, because they show where agency begins. Video and 3D systems are outside the selection but are mentioned where they inform the design of image-level agents.

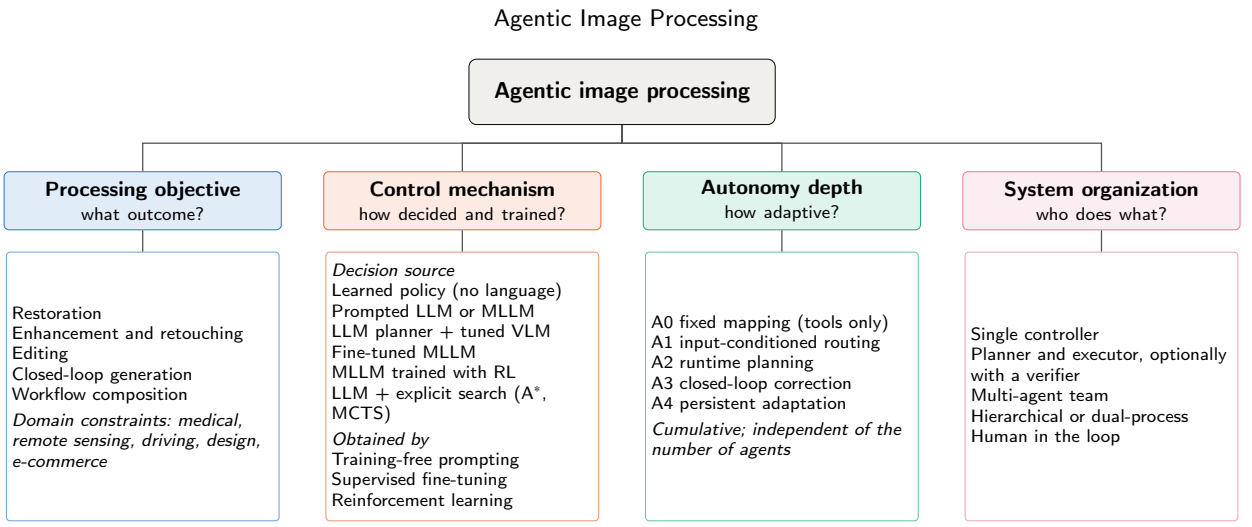


Figure 3: Four orthogonal axes for classifying agentic image-processing systems.

3.4. Characterization of the systems

For every selected system we recorded its task family and application domain, the type of controller and how it was obtained (training-free prompting, supervised fine-tuning, or reinforcement and preference learning), what the controller observes, which actions it can take, what feedback and memory it uses, how responsibilities are divided among its components, and its autonomy level. This characterization is based on the method description of each paper, and on the full text wherever finer details were needed. Quantitative results quoted in this review are taken from the original papers rather than from earlier surveys. A system is marked as peer-reviewed when a conference or journal version exists and as a preprint otherwise. The complete list of systems with their characterization is provided as supplementary material.

4. A four-axis taxonomy

Existing categorizations often mix unlike concepts in one hierarchy. "Restoration agents", "multi-agent systems", and "RL-based agents", for example, are sometimes listed as siblings, although a multi-agent restoration system trained with reinforcement learning belongs to all three. We therefore separate four orthogonal axes (Fig. 3). The *processing objective* describes what image outcome is sought, the *control mechanism* describes how decisions are made and how the controller is trained, the *autonomy depth* describes how much runtime adaptation is supported, and the *system organization* describes how responsibilities are allocated. Every system takes one value on each axis, and Fig. 4 shows how the reviewed systems are distributed.

4.1. Processing objective

Restoration recovers a latent clean image from degraded observations. It covers denoising, deblurring, deraining, dehazing, low-light correction, removal of compression artifacts, super-resolution, and mixtures of these. A reference image often exists in benchmarks but not at inference time, so the controller must estimate the degradation and its own progress from the image alone. *Enhancement and retouching* improve perceptual or aesthetic qualities such as exposure, color, tone, and local emphasis, for which no unique target exists, often through the parameters of professional tools. *Editing* changes selected content according to an instruction while preserving everything else, including multi-turn, compositional, layered, and design-document editing. *Closed-loop generation* synthesizes images from text or reference images and is included only when visual evidence obtained at run time changes a later generation or editing action. *Workflow composition* produces an executable image-processing program, typically a ComfyUI graph, whose execution yields the image. Application domains such as medical imaging, remote sensing, autonomous driving, design, and e-commerce (Section 7) are not separate objectives, but they impose constraints on fidelity, risk, and provenance.

Table 2
Autonomy depth: definitions, the behavioral evidence required to assign each level, the number of reviewed systems at each level, and examples.

Level	Definition	Required evidence	n	Examples
A0	Fixed mapping or predetermined pipeline	The same computation for every input, adapted only through conditioning	–	PromptIR, InstructIR (tools)
A1	Input-conditioned selection	An observation selects a model, path, workflow, or parameters once, without revision	8	DiffAgent, Derain-Agent, GenRouter, FlowTool
A2	Runtime planning	The goal is decomposed into a sequence of operations of variable length chosen at run time	23	JarvisArt, JarvisIR, IEAP, ImageEdit-R1
A3	Closed-loop correction	An observation after an action can accept, retry, roll back, replan, select among candidates, or stop	67	AgenticIR, 4KAgent, GenArtist, MIRA, CoSTA*
A4	Persistent adaptation	Experience, skills, tool profiles, or evaluators are updated across tasks and shown to change later decisions	10	FaSTA*, SEAR, SkillIR, PaAgent, JarvisEvo

4.2. Control mechanism

The controller is the component that implements Eq. (2). We distinguish six families by how decisions are produced.

(i) *Learned policies* without language, such as convolutional or other lightweight networks trained by reinforcement or supervised learning (Yu et al., 2018; Lu et al., 2026).

(ii) *Prompted LLMs or MLLMs* that plan from descriptions of the tools given in the prompt (Wang et al., 2024c).

(iii) *Hybrids of an LLM and a vision–language model (VLM)*, in which a text-only model plans and a separately tuned VLM perceives and verifies (Zhu et al., 2025b; Jiang et al., 2026b).

(iv) *Fine-tuned MLLMs* that map images directly to plans or tool calls (Chen et al., 2024; Lin et al., 2025a).

(v) *MLLMs trained with reinforcement learning* on the outcomes of executed actions (Lin et al., 2025b; Jia et al., 2026; Zeng et al., 2026).

(vi) *Language models combined with explicit search*, in which symbolic or Monte Carlo search operates over candidates proposed by the model (Gupta et al., 2025; Cui et al., 2026; Yao et al., 2026).

Independently of the family, a controller is obtained in one of three ways: by prompting without training, by supervised fine-tuning (SFT), often on trajectories found by an exhaustive offline search, or by reinforcement and preference learning. Training-free prompting is still the most common route, but Fig. 4(c) shows that controllers trained with reinforcement learning, rare in 2024, have become almost as common in 2026.

The key conceptual boundary within this axis lies between *conditioning* and *control*. A degradation prompt or a text embedding modulates a network without exposing a choice. Control requires a selection among alternatives with a meaningful counterfactual. FAPE-IR, for instance, uses an MLLM planner whose frequency-aware plan activates low-rank adaptation (LoRA) experts inside a diffusion model (Liu et al., 2025a), and RIRF uses a reasoning MLLM whose diagnosis conditions a restoration network (Yan et al., 2026). Both expose interpretable plans but decide only once, so we assign them to level A1.

4.3. Autonomy depth

Autonomy depth describes the strongest form of runtime adaptation for which a paper provides behavioral evidence (Table 2). It is a descriptive label, not a quality ranking. Higher levels add opportunities for evaluator error, latency, and irreproducibility, and an A1 router can be the right design for a narrow, safety-critical task. The levels are cumulative, so an A4 system also plans and verifies, and they are deliberately independent of the number of agents.

Two conventions matter when assigning levels. First, a learned policy that observes each intermediate state and can choose the next action or stop is closed-loop (A3) even if it never "reflects" in language. RL-Restore and Restore-R1 are examples (Yu et al., 2018; Lu et al., 2026). Second, persistence (A4) requires that stored experience change decisions on new inputs. Keeping the best image of the current episode, as the memory bank of CCA does (Hang et al., 2025), is memory within an episode and therefore counts as A3. Under these rules, closed-loop correction is by far the most common level, whereas persistent adaptation is demonstrated by only a few systems, all of them released in 2025 or 2026.

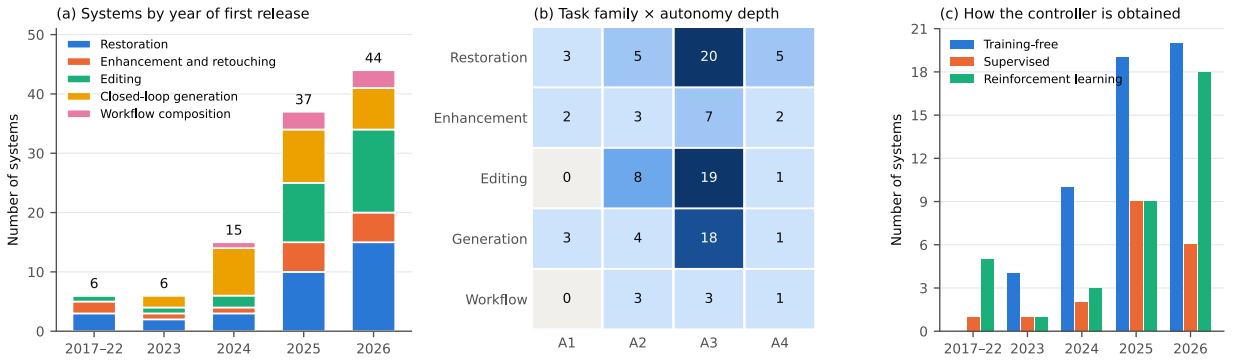


Figure 4: Distribution of the 108 reviewed systems. (a) Systems by year of first public release and task family (the 2026 column covers a partial year). (b) Task family versus autonomy depth. (c) How the controller is obtained, by period.

4.4. System organization

Organization describes how perception, planning, execution, and verification are allocated. A *single controller* makes all decisions. A *planner and executor*, optionally joined by a verifier, separate proposal, action, and assessment. A *multi-agent team* assigns roles that communicate through messages or a shared blackboard. *Hierarchical* and *dual-process* designs combine a fast path with a slow deliberative path (Li et al., 2025a; Gupta et al., 2026; Cui et al., 2026), and *human-in-the-loop* designs make users part of the loop through clarification questions, masks, or approvals (Hahn et al., 2025; Vinker et al., 2025). Organization is orthogonal to autonomy. The multi-agent team of MAIR narrows its search with a prior on the order of degradations and with one expert per degradation (Jiang et al., 2026b), whereas a single MLLM such as MIRA can implement a strong closed loop (Zeng et al., 2026). Several systems labelled multi-agent implement a fixed sequence of role prompts without feedback, and conversely several single-controller systems revise their trajectories extensively. A multi-agent organization should therefore be justified by measured gains over a single controller with the same tools, which few papers report (Section 9).

4.5. Trends

Three patterns stand out in Fig. 4. First, the field is young and growing quickly. Three quarters of the reviewed systems appeared in 2025 or 2026, and almost half are still preprints. This growth follows the development of MLLMs that can inspect an image, reason about a request, and emit structured tool calls, which removed the need to train a dedicated perception module and a dedicated planner for every task. Second, restoration and editing are the largest task families, but the mechanisms are converging. Closed-loop correction is the most common design in every family, and the same building blocks, namely diagnosis by an MLLM, structured tool descriptions, candidate selection by a judge, and rollback, recur across tasks. Third, the way controllers are obtained is changing. Early systems built on LLMs orchestrated proprietary models without any training. Since 2025, controllers are increasingly fine-tuned on trajectories produced by offline search and then optimized by reinforcement learning against rewards from image-quality metrics or MLLM judges. This shift raises the importance of reward design and judge reliability, which we analyze in Sections 5.8 and 8.

5. Architectural building blocks

Despite different task framings, the reviewed systems decompose into the same components (Fig. 1): intent understanding, diagnosis of the visual state, planning and search, tool representation and selection, execution, verification and stopping, memory, and the procedure by which the controller is trained. Table 3 summarizes design options and characteristic failure modes, and this section analyzes each component in turn.

5.1. Intent understanding and profiling

Restoration agents usually receive an implicit goal (“make it clean”) and must decide what clean means. 4KAgent makes this explicit through a profile module that fixes the target resolution, the handling of faces, the speed, and

Table 3
Components of an agentic image-processing system: design options, representative systems, and characteristic failure modes.

Component	Design options	Representative systems	Characteristic failure modes
Intent and profiling	Free-form instruction Structured profile (fidelity or perception, scale, faces) Clarification dialogue Region hints	4KAgent, proactive text-to-image agents, EyeControl	Underspecified goals filled with model defaults Too many clarification questions
Visual diagnosis	Learned state Degradation classifier Ensemble of NR-IQA metrics Description by a tuned VLM or MLLM Evidence from tool outputs	AgenticIR, Q-Agent, IAMAgent, EORestore-Agent	Confusion of visually similar degradations Hallucinated defects Blindness to accumulated artifacts
Planning and search	One-shot plan Greedy step-wise selection Depth-first search with rollback Tree of candidates A* search over a tool graph Monte Carlo tree search Learned policy	RestoreAgent, Q-Agent, GenArtist, CoSTA*, SEAR, PhotoAgent	Errors in operation order Combinatorial cost Myopic greedy choices Search that exploits judge errors
Tool representation	Text descriptions Tool cards and registries Tables of benchmarked capabilities Dependency graphs Standard tool protocols Mined subroutines	MAIR, CoSTA*, PerfGuard, Agent Banana, FaSTA*	Outdated or wrong capability descriptions Format errors in tool calls Difficulty adding new tools
Execution	Specialist networks Generative editors Professional software Classical operators Internal expert modules Code and workflow graphs	AgenticIR, JarvisArt, PhotoArtAgent, IEAP, ComfyAgent	Unrequested changes Loss from repeated resampling Non-determinism Interactions between tools
Verification and stopping	Binary success check Scalar quality gain Selection among candidates Pairwise judge Rubric Compilation or execution check Human approval	4KAgent, EORestore-Agent, Maestro, DiagramAgent	Judge bias and self-preference Reward hacking Stopping too early or never
Memory	History of the current episode Best-so-far buffer Experience documents Tool portrait banks Skill libraries Causal graphs Trained evaluator	AgenticIR, PaAgent, SkillIR, Causal-AgentIR, JarvisEvo	Contaminated memory Outdated experience Retrieval of irrelevant cases
Controller training	Prompting without training SFT on searched trajectories RL with IQA or MLLM rewards Preference learning Offline RL	RestoreAgent, TIR-Agent, MIRA, RetouchIQ	Overfitting to synthetic degradations Reward over-optimization Bias inherited from the teacher

the preference between fidelity and perceptual quality before execution (Zuo et al., 2025). The user's choice thereby replaces an arbitrary operating point on the perception–distortion curve (Blau and Michaeli, 2018). Retouching and editing agents receive instructions that are often aesthetic and underspecified. PhotoArtAgent first performs an explicit artistic analysis of subject, composition, and mood before planning (Chen et al., 2025b). PerTouch distinguishes strong

from weak instructions and maintains a scene-aware memory of user preferences (Chang et al., 2026), and EyeControl converts a few clicks or strokes into a structured description of where the viewer's attention should go (Qin et al., 2026). Proactive text-to-image agents go further. They maintain an explicit belief graph over entities and attributes and ask clarification questions where the importance-weighted uncertainty is highest (Hahn et al., 2025). Work on adaptive task reformulation shows that many editing failures are in fact formulation failures, caused by small targets, implicit spatial relations, or underspecified requests, and that rewriting the task into operations the editor handles well yields large gains without changing the editor (Zhao et al., 2026a).

5.2. Visual state diagnosis

The observation o_t is the agent's model of what is wrong or what remains to be done. Four sources are used, often in combination. *Learned states* feed a policy directly. Restore-R1, for example, concatenates frozen CLIP image features with a record of past actions (Lu et al., 2026). *Classifiers and IQA ensembles* provide calibrated but narrow signals. IAMAgent fine-tunes CLIP on a set of 15 degradation categories, raising top-1 classification accuracy from 38.27% to 94.57% (Wei et al., 2026), while Q-Agent and 4KAgent aggregate several NR-IQA metrics (Zhou et al., 2025; Zuo et al., 2025). *Tuned vision–language models* produce graded, compositional diagnoses. AgenticIR adapts DepictQA with LoRA to detect degradations and judge intermediate results (Zhu et al., 2025b), and Q-Agent decomposes the perception of multiple degradations into one chain-of-thought question per degradation (Zhou et al., 2025). Finally, *evidence from tool outputs*, such as masks and boxes from open-set detectors and segmenters (Liu et al., 2024b; Kirillov et al., 2023), optical character recognition, and layer decompositions, grounds editing decisions (Wang et al., 2024c; Yu et al., 2026b; Yun et al., 2026).

Diagnosis is the first point of failure. The tuned perceiver of AgenticIR reaches an F1 score of 0.95 for noise and 0.98 for rain, but only 0.65 for motion blur and 0.72 for defocus blur, which the authors attribute to the visual similarity of the two blur types (Zhu et al., 2025b). Such errors propagate. A misdiagnosed degradation selects the wrong tool, and a judge with the same blind spot will accept the result. SkillIR makes the dynamic nature of diagnosis explicit. It observes that each accepted tool execution changes the remaining degradation and therefore the applicability of later tools, and it reassesses the remaining degradations after every committed action (Shao et al., 2026).

5.3. Planning and search

Planning determines the sequence and parameters of actions. Six strategies appear, each trading computation for robustness.

One-shot programs. The MLLM of RestoreAgent emits an ordered program of tasks and models, learned from an exhaustive offline search (Chen et al., 2024). JarvisArt emits a complete Lightroom configuration (Lin et al., 2025b), and DiTTo is trained on optimal restoration trajectories generated by a simulator (Choi and Oh, 2026). One-shot plans are fast and auditable, but they cannot react to execution failures unless the plan is regenerated.

Greedy step-wise selection. Q-Agent applies every currently applicable tool, scores the candidates with an NR-IQA ensemble, commits the best one, and repeats (Zhou et al., 2025). EORestore-Agent accepts a step only if a learned scorer predicts no deterioration in PSNR, structural similarity (SSIM), or learned perceptual image patch similarity (LPIPS) and a positive gain in PSNR, and otherwise keeps the current image (Qi et al., 2026). Greedy control is simple and conservative but myopic.

Depth-first search with rollback. AgenticIR executes one operation, lets the perceiver judge it, rolls back failures, and reschedules, which amounts to a guided depth-first search over restoration paths (Zhu et al., 2025b). The approach is robust but expensive. Restore-R1 reports AgenticIR runtimes of 154–345 minutes over its four evaluation settings, compared with 80–122 minutes for its own learned policy (Lu et al., 2026).

Prior-constrained planning. MAIR imposes a prior on the order in which real-world degradations arise (first the scene, then the imaging process, then compression) and undoes them in reverse order, which shrinks the search space before any tool is invoked (Jiang et al., 2026b).

Explicit search over tool graphs. CoSTA* lets a language model build a tree of subtasks and then runs cost-sensitive A* search over a graph of tool dependencies, with a heuristic that combines benchmarked quality and runtime. Vision–language checks of executed steps update the path costs (Gupta et al., 2025). GenArtist explores a planning tree with alternative recovery branches (Wang et al., 2024c), and MSRAMIE organizes intermediate states and reference images as a tree of states and a graph of references (Qiu et al., 2026).

Monte Carlo tree search (MCTS). SEAR plans with MCTS that prunes unpromising branches, guided by a hybrid no-reference reward and a tournament among MLLM judges designed to prevent the exploitation of any single metric

(Cui et al., 2026). PhotoAgent simulates candidate edits at reduced resolution with MCTS, executes only the top-k actions at full resolution, and rolls back if no candidate improves the image (Yao et al., 2026). Both echo the classical combination of learned priors with tree search (Silver et al., 2016).

Learned policies (Section 5.8) replace repeated search with a single forward pass. TIR-Agent reports an inference speed-up of more than 2.5× from eliminating redundant tool executions (Jia et al., 2026), and EvoIR-Agent reports a 61% gain in efficiency from a self-evolving pool of experience (Zhuang et al., 2026). Deeper planning is not always better. The average trajectory length of MIRA grows only from 2.98 to 4.21 steps as the step budget rises from three to seven, and its overall editing score peaks at the three-step setting (Zeng et al., 2026).

5.4. Tool representation, selection, and parameter control

A controller can only use what it understands. Early systems described tools in natural language within the prompt (Wu et al., 2023a; Shen et al., 2023), but structured representations now dominate. CCA describes each tool at two levels, with short descriptions for planning and detailed manuals for execution, which reduced incorrectly formatted tool calls from over 20% to under 10% (Hang et al., 2025). The registry cards of MAIR record the function, the applicable severity, the speed, and the side effects of each tool, so that new tools can be added without retraining (Jiang et al., 2026b). CoSTA* maintains a table of model descriptions, a graph of tool dependencies, and a benchmark table of quality and runtime priors (Gupta et al., 2025). PerfGuard treats the competence of a tool as an empirical, multidimensional profile that is updated whenever executed results disagree with the predicted ranking (Chen et al., 2026c). PaAgent builds a "portrait bank" summarizing how each restoration tool behaved on past inputs and retrieves it when selecting tools (Wang et al., 2026). Agent Banana exposes its editing tools through a server that follows the Model Context Protocol (MCP), a common standard for connecting language models to external tools (Ye et al., 2026).

Selection and parameterization interact. Retouching agents choose continuous parameters of professional tools. JarvisArt and RetouchIQ emit Lightroom adjustments (Lin et al., 2025b; Wu et al., 2026), and IEA operates 16 parameterized tools step by step (Zhu et al., 2026b). FlowTool replaces the token-by-token generation of parameters with a flow-matching model that generates all tool parameters at once, and reports at least a 50-fold reduction in latency relative to MLLM editing agents (Le et al., 2026). Derain-Agent predicts, in addition to a sequence of tools, a pixel-wise strength map for each tool, showing that spatially varying strength is a decision variable in its own right (Yu et al., 2026c). Changing the set of actions can matter more than improving the planner. A tool trained on mixed degradations raises the average PSNR of HybridAgent over 14 mixed settings from 19.84 to 24.83 dB compared with chaining tools trained on single degradations (Li et al., 2025a), and OPERA trains its tools jointly so that they cooperate when applied in sequence (Zhu et al., 2026a).

5.5. Execution interfaces

Execution determines what can go wrong after a correct decision. *Specialist networks* such as restoration models are deterministic and well characterized, but chaining them compounds artifacts, and their interactions are hard to predict (Zhu et al., 2025b, 2026a). *Generative editors* are flexible but can regenerate content they were not asked to change, hallucinate objects, or drift in identity over multiple turns. In a study of shadow removal, a commercial editor produced clean results but could also "misread a shadow as material or geometry", which motivated generating several candidates and selecting among them with physics-based checks (Hu et al., 2026). *Professional software*, such as Lightroom or classical image operators, yields interpretable, reversible, and resolution-independent edits that preserve content by construction (Chen et al., 2025b; Lin et al., 2025b; Yao et al., 2026). *Localized execution* limits unrequested changes. Agent Banana crops the target layer at native resolution, edits it in local coordinates, and blends it back, so pixels outside the target remain untouched (Ye et al., 2026), and I2E decomposes an image into object layers on which a vision–language–action agent performs atomic actions (Yu et al., 2026b). *Foundation-model executors* invert the usual design. HarnessIR argues that real mixed degradations cannot be undone one at a time. It instead composes the diagnosis and the collected evidence into a prompt for a multimodal foundation model that restores the image in a single pass, followed by refinement driven by verification (Kong et al., 2026). *Programs and workflows*, such as drawing code for diagrams or typed ComfyUI graphs, make the output editable and allow a compiler or interpreter to check the execution (Wei et al., 2025; Xue et al., 2025; Li et al., 2026b).

5.6. Verification, rollback, and stopping

Verification turns planning into closed-loop control. Six kinds of verifiers appear in the reviewed systems:

(i) Binary success checks by a tuned VLM (Zhu et al., 2025b; Li et al., 2025a).

(ii) Scalar quality differences computed by NR-IQA ensembles, human-preference models, or learned relative scorers (Zhou et al., 2025; Zuo et al., 2025; Qi et al., 2026).

(iii) Selection among several tool outputs. In the quality-driven mixture of experts of 4KAgent, for instance, all eligible experts are executed for a step and the best result is kept (Zuo et al., 2025).

(iv) MLLM critics producing rubric scores or pairwise verdicts (Venkatesh et al., 2025; Wan et al., 2025; Ku et al., 2024a).

(v) Deterministic execution checks, such as compiling generated diagram code before semantic inspection (Wei et al., 2025) or reverting invalid workflow edits (Li et al., 2026b).

(vi) Human approval (Wei et al., 2026).

Ablations confirm that verification matters. Removing reflection lowers the PSNR of AgenticIR in its group-A setting from 21.12 to 20.47 dB (Zhu et al., 2025b), and the compositional score of GenArtist rises from 0.3739 with tool selection alone to 0.4095 with a verified planning chain and to 0.4499 with a planning tree (Wang et al., 2024c). Yet verifiers are also the weakest link. EORestore-Agent shows that replacing its learned scorer, which is trained with full-reference supervision, by differences of no-reference quality scores costs 1.1–4.6 dB on remote-sensing composites. It also shows that agents built for natural images, applied without adaptation, fell below the degraded input in PSNR in 17 of 18 settings (Qi et al., 2026). Banana100 shows that none of 21 popular NR-IQA metrics consistently scores heavily degraded, iteratively edited images below clean ones (Tang et al., 2026). Restore-R1, which uses an MLLM quality score as its training reward, improves no-reference metrics while full-reference fidelity declines as more actions are taken. This is the perception–distortion trade-off reappearing as reward over-optimization (Lu et al., 2026). Self-refinement can also exploit the weaknesses of an evaluator when generator and judge share a model (Pan et al., 2024; Xu et al., 2024). Robust designs therefore separate the judge from the actor, combine heterogeneous judges, anchor them to human labels as JarvisEvo does for its evaluator (Lin et al., 2025c), and prefer conservative acceptance rules.

Stopping is the least studied decision. Explicit stop actions appear in reinforcement-learning restorers (Yu et al., 2018) and in the training data of RestoreAgent, which includes completed states that require termination (Chen et al., 2024). Early stopping in CCA cut the average execution from 20 tool calls over five rounds to 12 calls over about three rounds on 20 requests (Hang et al., 2025), and PhotoAgent stops when no candidate improves the image any further (Yao et al., 2026). Most systems still rely on a fixed budget of steps.

5.7. Memory and self-evolution

Memory distinguishes A3 from A4 behavior. *Memory within an episode* includes the trajectory history, a buffer holding the best result so far, and the dialogue state (Hang et al., 2025; Wei et al., 2023). Agent Banana's context folding keeps image states as nodes of a state graph, retains the context of each tool call only temporarily, and stores verified editing paths for later steps (Ye et al., 2026). *Experience across tasks* takes several forms. AgenticIR produces experience documents offline by executing all tool orders on example mixtures and summarizing which orders succeed (Zhu et al., 2025b). EvoIR-Agent maintains a hierarchical experience pool that evolves from accumulated records (Zhuang et al., 2026), and PaAgent keeps its tool portrait bank (Wang et al., 2026). Causal-AgentIR builds a causal memory graph that links degradations, regions, tools, quality changes, and preferences, with explicit operations to add, merge, reinforce, or discard experience (Gao et al., 2026). SEAR keeps an episodic memory indexed by degradation-aware descriptions of the image state, which avoids repeating expensive searches (Cui et al., 2026). *Skills* compress experience into reusable procedures. FaSTA* mines subroutines from execution traces, accepts a new rule only when it yields a net benefit, and falls back to localized A* search when no subroutine applies (Gupta et al., 2026). SkillIR represents experience as scene-aware skills that describe their conditions of use, expected effects, and known failures (Shao et al., 2026), and ComfyClaw distils trajectories and verifier feedback into a skill library that is revealed to the controller step by step (Li et al., 2026b). Finally, *trained evaluators* are memory in parametric form. JarvisEvo trains its editor and its evaluator together while masking the tokens of its own scores to prevent reward leakage (Lin et al., 2025c).

Persistent memory raises questions that the papers rarely answer: whether experience transfers across tool versions and degradation distributions, how contaminated or outdated entries are detected, and whether reusing memory makes outputs more uniform. Causal-AgentIR is one of the few systems that treats forgetting and revision as explicit operations (Gao et al., 2026).

5.8. How the controller is trained

Training-free prompting is the fastest route to a working agent and makes tools easy to swap. It inherits, however, the cost, latency, and version changes of proprietary models. IAMAgent, for instance, acknowledges its dependence on ChatGPT, on prompt engineering, and on token limits (Wei et al., 2026). *Supervised fine-tuning* compresses expensive search into a model. RestoreAgent and Derain-Agent label each training image with the best pipeline found by exhaustive offline search (Chen et al., 2024; Yu et al., 2026c), and DiTTo reduces the cost of building such trajectory datasets with a learned simulator (Choi and Oh, 2026). *Reinforcement learning* optimizes the outcomes of executed actions. Restore-R1 trains an actor–critic policy with 0.28 billion parameters using rewards from an MLLM quality judge, without clean reference images (Lu et al., 2026). TIR-Agent combines SFT with RL, perturbs the SFT trajectories to broaden exploration, and re-weights heterogeneous IQA rewards to mitigate reward hacking (Jia et al., 2026). PaAgent combines a subjective MLLM assessment with seven NR-IQA metrics in its reward (Wang et al., 2026), and OPERA optimizes the composition of tools over a combinatorial space of plans with the final restoration quality as reward (Zhu et al., 2026a).

In editing and retouching, MIRA applies group relative policy optimization (GRPO) (Shao et al., 2024) with a composite editing reward. Compared with SFT alone, GRPO improves its GPT-judged semantic consistency by 14.85% with one editor (Zeng et al., 2026). Retouching agents optimize their plans with different rewards. JarvisArt rewards correct format, correct tools, and visual similarity of the executed result (Lin et al., 2025b). RetouchIQ uses a generalist MLLM reward model that generates case-specific metrics (Wu et al., 2026), and IEA rewards improvements in likeness, the usefulness of tools, and the summarization of intent (Zhu et al., 2026b). PosterCraft trains a poster generator with best-of-n preference optimization and then fine-tunes a VLM to write content and style suggestions for each generated poster, which the generator uses to produce a refined version (Chen et al., 2025c). Offline RL from quality-scored trajectories trains styling planners (Mukherjee et al., 2026), and preference optimization aligns routers and planners (Zhao et al., 2024; Chen et al., 2026c).

The progression from prompting to SFT to RL mirrors research on language agents (Ouyang et al., 2022; Rafailov et al., 2023), with one crucial difference. Rewards in image processing are perceptual and largely unverifiable at inference time. Every RL-trained agent reviewed here optimizes a proxy, whether an NR-IQA metric, a preference model, or an MLLM judge, so its improvements are only as trustworthy as that proxy.

5.9. Multi-agent and human collaboration

Multi-agent designs assign perception, planning, execution, and critique to separate roles. The reported benefits are real but modest and rarely isolated from confounding factors. Users of CCA preferred the full model, which combines collaboration and competition among agents, in 63% of comparisons against a variant without collaboration (Hang et al., 2025), and a multi-agent framework for amodal completion improved LPIPS from 0.221 to 0.127 over its single-agent baseline (Fan et al., 2025). Other designs rely on competition and consensus. In CREA, a critic and a director may disagree before reaching consensus on a six-dimensional creativity score (Venkatesh et al., 2025), and Maestro uses MLLM critics with pairwise comparisons to evolve prompts (Wan et al., 2025). Few studies compare a team against a single controller with identical tools and token budget. Small language models solving mathematical problems with self-refinement, reflection, or debate are not reliably better than repeated sampling at equal token cost (Mirzaei, 2026), so such controls should become standard. Human collaboration ranges from approval and correction in conversational restoration (Wei et al., 2023, 2026) to shared canvases on which human and agent strokes live in one representation (Vinker et al., 2025) and belief graphs that users can inspect and edit (Hahn et al., 2025).

6. Task-level synthesis

This section compares systems within each task family. Tables 4–7 list representative systems with their controller, observation, actions, feedback and memory, and autonomy level. In the tables, P denotes a prompted controller without training, RL denotes reinforcement or preference learning, PR marks peer-reviewed papers, and pp marks preprints.

6.1. Restoration

Restoration is the most mature task family and the clearest test bed for agentic control, because the ground truth exists in benchmarks while the controller must act without it. Three generations are visible. *Search-based orchestration* systems such as AgenticIR, MAIR, and 4KAgent keep a large toolbox of specialist models and use a perceiver and a judge to explore orderings at run time (Zhu et al., 2025b; Jiang et al., 2026b; Zuo et al., 2025). They need no training

Table 4
Agentic restoration systems (ordered by first release).

System	Year	Controller (training)	Observation → actions	Feedback and memory	Level
RL-Restore (Yu et al., 2018)	2018 PR	CNN policy (RL)	learned state → small restoration CNNs + stop	per-step state; stop action	A3
PixelRL (Furuta et al., 2020)	2019 PR	per-pixel FCN policy (RL)	learned state → pixel-wise filters	iterative state	A3
Clarity ChatGPT (Wei et al., 2023)	2023 pp	ChatGPT (P)	CLIP degradation detector, NR-IQA → IR models, regions, fusion	user feedback; dialogue	A3
AutoDIR (Jiang et al., 2024)	2023 PR	blind IQA + latent diffusion (SFT)	dominant-degradation detection → diffusion restoration skill	iterate until no degradation	A3
RestoreAgent (Chen et al., 2024)	2024 PR	LLaVA-Llama3 (SFT on searched pipelines)	MLLM → ordered task–model program	re-observation, rollback, stop states	A3
Chain-of-Restoration (Cao et al., 2024)	2024 pp	degradation discriminator + multi-task model	per-step basis detection → one degradation removed per step	iterate until clean	A3
AgenticIR (Zhu et al., 2025b)	2024 PR	GPT-4 + LoRA-tuned DepictQA	VLM diagnosis → specialist toolbox (8 degradations)	reflection, rollback, rescheduling; experience documents	A3
MAIR (Jiang et al., 2026b)	2025 PR	GPT-4o scheduler + experts (P)	DepictQA → tools via registry cards	degradation-order prior; per-expert stop	A3
HybridAgent (Li et al., 2025a)	2025 pp	fast LLM + slow MLLM (SFT)	MLLM → LoRA tools for single and mixed degradations	feedback agent judging whether the image is clean	A3
JarvisIR (Lin et al., 2025a)	2025 PR	VLM (SFT + human-feedback alignment)	VLM → expert restoration models	–	A2
Q-Agent (Zhou et al., 2025)	2025 pp	Qwen2-VL (SFT, CoT)	per-degradation CoT → tools	greedy NR-IQA candidate selection	A3
4KAgent (Zuo et al., 2025)	2025 PR	LLM and VLM agents (P)	IQA + VLM → experts, including a face branch	quality-driven selection among experts, rollback	A3
FAPE-IR (Liu et al., 2025a)	2025 pp	frozen MLLM planner	frequency-aware plan → LoRA-MoE in diffusion executor	–	A1
Restore-R1 (Lu et al., 2026)	2025 PR	0.28B actor–critic (RL, MLLM reward)	CLIP state → toolbox	per-step state	A3
Derain-Agent (Yu et al., 2026c)	2026 pp	ResNet scheduler (SFT)	residual features → tool path + strength maps	–	A1
PaAgent (Wang et al., 2026)	2026 pp	Qwen3.5 (GRPO) + retrieval	MLLM + NR-IQA → 14 tools	self-evolving tool portrait bank	A4
TIR-Agent (Jia et al., 2026)	2026 PR	VLM (SFT + RL)	VLM → direct tool calls	multi-metric adaptive reward	A3
RIRF (Yan et al., 2026)	2026 pp	Qwen3-VL reasoner (SFT, RL)	structured diagnosis → conditioned restorer	–	A1
EvoIR-Agent (Zhuang et al., 2026)	2026 pp	MLLM (P)	VLM → tools and orders	hierarchical self-evolving experience pool	A4
OPERA (Zhu et al., 2026a)	2026 pp	planner (RL)	MLLM → co-trained tools	quality reward at training	A2
DiTTo (Choi and Oh, 2026)	2026 pp	VLM (SFT on simulator trajectories)	VLM → order-aware expert actions	–	A2
SEAR (Cui et al., 2026)	2026 PR	LLM + MCTS; intuitive executor	MLLM + NR-IQA → tools	MCTS, MLLM tournament; episodic memory	A4
Causal-AgentIR (Gao et al., 2026)	2026 pp	multi-agent LLM and VLM (P)	VLM → tools	critique; causal memory graph	A4
SkillIR (Shao et al., 2026)	2026 pp	MLLM (P)	residual-state assessment → one bounded action	transition verification; evolving skills	A4
HarnessIR (Kong et al., 2026)	2026 pp	MLLM harness (P)	diagnosis + evidence → foundation-model executor	verification-driven refinement	A3
EORestore-Agent (Qi et al., 2026)	2026 pp	tuned VLM + relative scorer	residual degradations → tool pools	step accepted only if predicted gain	A3

at the planning level and are flexible, but they pay for exploration, since rollback and reflection multiply tool calls and runtime grows with the number of degradations (Lu et al., 2026). *Learned planners* compress this search into a model. RestoreAgent and DiTTo learn from searched or simulated optimal trajectories (Chen et al., 2024; Choi and Oh, 2026), while Restore-R1, TIR-Agent, PaAgent, and OPERA learn policies with reinforcement learning (Lu et al., 2026; Jia et al., 2026; Wang et al., 2026; Zhu et al., 2026a). *Experience-driven* agents keep search for novel cases but reuse what worked before. EvoIR-Agent, SEAR, Causal-AgentIR, and SkillIR differ mainly in what they store, namely experience summaries, episodes indexed by image state, causal graphs, or conditional skills (Zhuang et al., 2026; Cui et al., 2026; Gao et al., 2026; Shao et al., 2026).

Three findings recur. First, *order and interaction effects are real*. The experience-guided schedules of AgenticIR outperform random tool invocation on most metrics, and the degradation-order prior of MAIR reduces search while

Table 5
Agentic enhancement and retouching systems.

System	Year	Controller (training)	Observation → actions	Feedback and memory	Level
Exposure (Hu et al., 2018)	2017 PR	policy (RL + adversarial)	learned state → differentiable filters	sequential; stop	A3
RL editing software (Kosugi and Yamasaki, 2020)	2019 PR	policy (RL, unpaired)	learned state → software operations	sequential	A3
ALL-E (Li et al., 2023)	2023 PR	policy (RL, aesthetic reward)	learned state → curve adjustments	iterative	A3
PhotoArtAgent (Chen et al., 2025b)	2025 pp	VLM (P)	artistic analysis, histograms → Lightroom parameters	self-critique and refinement	A3
JarvisArt (Lin et al., 2025b)	2025 PR	Qwen2.5-VL (SFT + GRPO)	MLLM → Lightroom tool configuration	–	A2
PerTouch (Chang et al., 2026)	2025 PR	VLM agent + diffusion retoucher	semantic regions → parameter maps	feedback-driven rethinking; preference memory	A4
JarvisEvo (Lin et al., 2025c)	2025 pp	MLLM editor–evaluator (SFT + RL)	interleaved image CoT → Lightroom calls	self-evaluation, reflection; co-evolved evaluator	A4
RetouchIQ (Wu et al., 2026)	2026 PR	MLLM (RL, generalist reward model)	MLLM → executable adjustments	–	A2
PhotoAgent (Yao et al., 2026)	2026 PR	VLM + MCTS	perceiver → classical and generative edits	top-k real execution; rollback	A3
IEA (Zhu et al., 2026b)	2026 pp	VLM (SFT + GRPO)	MLLM → 16 parameterized tools	step-wise refinement	A3
EyeControl (Qin et al., 2026)	2026 PR	MLLM + diffusion executor	clicks + image → focus-region tonal edits	–	A2
FlowTool (Le et al., 2026)	2026 pp	flow-matching parameter generator	VLM features → tool parameters	–	A1

preserving quality (Zhu et al., 2025b; Jiang et al., 2026b). Restore-R1 observes that a dehazing model can improve low-light images, so a learned sequence can beat a fixed mapping from degradations to tools (Lu et al., 2026). Second, *the tool set bounds the agent*. Tools trained on mixed degradations, tools trained jointly, and foundation-model executors yield gains that no amount of planning over tools for single degradations can match (Li et al., 2025a; Zhu et al., 2026a; Kong et al., 2026). Third, *full-reference and no-reference evidence often disagree*. Restore-R1 leads on no-reference metrics in unseen settings while AgenticIR retains a higher PSNR (Lu et al., 2026), and an agent can raise perceptual scores while lowering fidelity. The MiO100 benchmark of mixed degradations, introduced with AgenticIR, has become a common test set, but most evaluations remain synthetic, and evidence on real images is mostly qualitative (Zhu et al., 2025b; Kong et al., 2026). The same pattern extends beyond still images, as MoA-VR applies identification, routing, and quality-assessment agents to all-in-one video restoration (Liu et al., 2025b). Chain-of-Restoration and AutoDIR show that a step-by-step loop driven by observations can also live inside a learned model rather than an external agent (Cao et al., 2024; Jiang et al., 2024). Both remove one degradation per step and stop when the image is judged clean, which blurs the line between all-in-one restoration and agency.

6.2. Enhancement and retouching

Retouching differs from restoration in that there is no unique target, and the actions are usually continuous, interpretable parameters (Table 5). The lineage begins with reinforcement-learning policies such as Exposure, which applies a sequence of differentiable filters (Hu et al., 2018), the policy of Kosugi and Yamasaki (2020), which operates conventional editing software, and ALL-E, which adjusts tone curves under an aesthetic reward (Li et al., 2023). These designs make retouching agents unusually auditable. JarvisArt, RetouchIQ, IEA, and PhotoArtAgent all output parameter traces that a photographer can inspect and edit in Lightroom or a similar tool (Lin et al., 2025b; Wu et al., 2026; Zhu et al., 2026b; Chen et al., 2025b). The central difficulty is the reward. Rule-based rewards that compare the result with a single reference retouch penalize legitimate alternatives. RetouchIQ therefore trains a generalist reward model that generates case-specific metrics (Wu et al., 2026), PhotoAgent learns an aesthetic reward from a benchmark of 7,000 user-generated photos (Yao et al., 2026), and JarvisEvo trains its editor and evaluator together while anchoring the evaluator to human scores (Lin et al., 2025c). Personalization is emerging as a distinctive requirement. PerTouch stores long-term user preferences, and EyeControl interprets weak user intent about where attention should go (Chang et al., 2026; Qin et al., 2026). Finally, the need for speed is pushing parameter prediction away from token-by-token generation toward generative models that output all parameters at once (Le et al., 2026).

Table 6
Agentic editing systems.

System	Year	Controller (training)	Observation → actions	Feedback and memory	Level
CAISE (Kim et al., 2022)	2022 PR	dialogue model (SFT)	dialogue + image → editing commands	user turns	A2
Visual ChatGPT (Wu et al., 2023a)	2023 pp	ChatGPT (P)	captions and tools → visual foundation models	user turns	A2
CCA (Hang et al., 2025)	2024 PR	GPT-4 planners, VLM discriminator (P)	VQA, aesthetics → 20 tools	collaboration + competition; best-result buffer	A3
GenArtist (Wang et al., 2024c)	2024 PR	GPT-4V (P)	MLLM + detector → 10 generation, 8 editing tools	planning tree with verification	A3
CoSTA* (Gupta et al., 2025)	2025 pp	LLM + A* (P)	VLM checks → 24 tools on dependency graph	search-cost update	A3
FaSTA* (Gupta et al., 2026)	2025 PR	LLM + subroutines + A* (P)	VLM checks → subroutines or tools	mined subroutine rules (persistent)	A4
EmoAgent (Mao et al., 2026)	2025 PR	planning, editing, and critic agents (P)	critic → diverse affective edits	critic refinement	A3
IEAP (Hu et al., 2025)	2025 PR	VLM programmer (P)	instruction → atomic adapters of a diffusion transformer	–	A2
RefineEdit-Agent (Liang et al., 2025)	2025 pp	LLM planner + VLM evaluator (P)	scene parsing → editing tools	VLM evaluation loop	A3
MIRA (Zeng et al., 2026)	2025 PR	Qwen2.5-VL (SFT + GRPO)	current image → atomic edit prompt or stop	per-step re-observation	A3
Multi-agent amodal (Fan et al., 2025)	2025 PR	MLLM roles (P)	occlusion analysis → masks, inpainting	–	A2
CEARI (Song et al., 2025)	2025 PR	evolutionary hyper-heuristic agents	shared evidence → swaps, inpainting	alternating optimization	A3
I2E (Yu et al., 2026b)	2026 PR	vision–language–action agent (SFT)	object layers → atomic layer actions	–	A2
Agent Banana (Ye et al., 2026)	2026 pp	planner + executor (P)	MLLM → MCP tools applied to cropped layers	verification; context folding	A3
ImageEdit-R1 (Zhao et al., 2026c)	2026 pp	decomposition VLM (GRPO)	instruction → action–subject–goal sub-requests	–	A2
MSRAMIE (Qiu et al., 2026)	2026 pp	MLLM instructor (P)	intermediate states → editor prompts	tree of states, graph of references	A3
IMAGAgent (Shen et al., 2026a)	2026 pp	VLM planner + LLM reflector (P)	current image → retrieval, segmentation, editing	multi-expert reflection	A3
CAMEO (Pu et al., 2026)	2026 pp	multi-agent MLLM (P)	evaluation → editors + adaptive references	iterative refinement	A3
Task reformulation (Zhao et al., 2026a)	2026 pp	MLLM (P)	analysis → reformulated sub-operations	feedback refinement	A3
EditRefiner (Xu et al., 2026a)	2026 pp	perception, reasoning and action agents (SFT)	artifact saliency → local corrections	evaluation loop	A3
Plans-to-Pixels (Rajan et al., 2026)	2026 pp	planner + orchestrator (RL)	VLM judge → tools and regions	outcome reward	A3
GMO-E^{2}DIT (Guo et al., 2026)	2026 pp	VLM (SFT)	region-grounded agenda → mask-conditioned editor	reflection, retry, safe partial progress	A3
ReDesign (Yun et al., 2026)	2026 PR	MLLM (P)	tool outputs → layer-hierarchy expansion	accept, prune, or retry	A3
MiLDEAgent (Lin et al., 2026b)	2026 pp	multimodal reasoner (RL)	layer analysis → layer editor	–	A2

6.3. Editing

Editing agents (Table 6) face a dual objective, namely to satisfy the instruction and to preserve everything else. Early conversational systems such as CAISE and Visual ChatGPT mapped dialogue turns to editing commands and left verification to the user (Kim et al., 2022; Wu et al., 2023a). Later work has explored three answers. *Decomposition* turns complex instructions into atomic operations. IEAP programs a sequence of atomic adapters, ImageEdit-R1 trains a decomposition policy that outputs actions, subjects, and goals, and MIRA predicts one atomic edit at a time from the current image (Hu et al., 2025; Zhao et al., 2026c; Zeng et al., 2026). On complex instructions, MIRA raised the GPT-judged semantic consistency of Qwen-Image-Edit from 6.29 to 6.88 (Zeng et al., 2026). *Tool orchestration with search* chooses among heterogeneous tools, such as detectors, segmenters, inpainters, and generators, and verifies each step (Hang et al., 2025; Wang et al., 2024c; Gupta et al., 2025; Shen et al., 2026a). On MagicBrush, GenArtist reported a CLIP image similarity (CLIP-I) of 0.9403 for single-turn and 0.9071 for multi-turn editing (Wang et al., 2024c). Several systems use dedicated critics for this verification. EmoAgent refines affective edits with a critic agent (Mao et al., 2026), CAMEO combines several editors with adaptive references under iterative evaluation (Pu et al., 2026), and EditRefiner locates artifacts in an edited image and corrects them locally (Xu et al., 2026a). Plans-to-Pixels

Table 7
Closed-loop generation and workflow-composition systems (a selection).

System	Year	Controller (training)	Observation → actions	Feedback and memory	Level
Idea2Img (Yang et al., 2024b)	2023 PR	GPT-4V (P)	draft images → text-to-image prompts	iterative self-refinement	A3
SLD (Wu et al., 2024d)	2023 PR	LLM (P) + detector	detected objects → adding, removing, or moving objects in the latent	self-correction	A3
CompAgent (Wang et al., 2024d)	2024 pp	LLM (P)	layout + VQA → composition tools	verification, human feedback	A3
MuLan (Li et al., 2024b)	2024 pp	LLM + VLM (P)	per-object VLM check → stage-wise diffusion	regenerate on violation	A3
DiffAgent (Zhao et al., 2024)	2024 PR	LLaMA-2 (SFT + ranking)	prompt → model API + hyperparameters	–	A1
ComfyAgent (Xue et al., 2025)	2024 PR	multi-agent LLM (P)	workflow execution → workflow code	execution feedback	A3
Proactive agents (Hahn et al., 2025)	2024 PR	LLM (P)	belief graph → clarification questions	user answers	A3
T2I-Copilot (Chen et al., 2025a)	2025 PR	interpreter, engine, evaluator (P)	quality report → model choice, prompts, regions	evaluation loop	A3
ComfyGPT (Huang et al., 2025b)	2025 pp	four agents (SFT + RL)	validation → node connections	refinement	A3
Maestro (Wan et al., 2025)	2025 pp	MLLM critics + verifier (P)	critiques → evolved prompts	pairwise judging	A3
ImAgent (Wang et al., 2025a)	2025 pp	unified MLLM (P)	self-evaluation → generation actions	policy-controlled test-time scaling	A3
PosterCraft (Chen et al., 2025c)	2025 pp	generator + VLM critic (SFT + RL)	poster critique → feedback-conditioned regeneration	content and style suggestions	A3
Agentic Retoucher (Shen et al., 2026b)	2026 PR	perception, reasoning and action agents	distortion saliency → local inpainting	perception–reasoning–action loop	A3
GenAgent (Jiang et al., 2026a)	2026 pp	MLLM (SFT + agentic RL)	judgement → text-to-image tool calls	reflection	A3
PerfGuard (Chen et al., 2026c)	2026 pp	analyst, planner, worker (preference optimization)	self-evaluation → ranked tools	updated capability matrix	A4
GenRouter (Chen et al., 2026a)	2026 pp	router	demand profile → workflow template	experience matching	A1
ComfyClaw (Li et al., 2026b)	2026 pp	LLM (P)	region VLM verifier → typed graph edits	revert + repair; skill library	A4
WeAgent-MMGenEdit (Zhang et al., 2026)	2026 pp	MLLM (SFT + RL)	retrieved evidence → generator or editor	verification tools	A3

trains its planner and orchestrator with an outcome reward from a VLM judge (Rajan et al., 2026), and CEARI lets co-evolving agents reassemble fragmented images and inpaint their missing pieces (Song et al., 2025). *Preservation by construction* restricts edits to regions or layers. Agent Banana edits cropped layers at native resolution, GMO-E^{2}DIT executes sub-programs conditioned on masks and retains safe partial progress, and I2E converts pixels into object layers that can be manipulated (Ye et al., 2026; Guo et al., 2026; Yu et al., 2026b).

Multi-turn editing exposes failure modes that single-turn benchmarks hide. Error accumulation and semantic drift motivate the constraint-aware planning of IMAGAgent (Shen et al., 2026a). Iterative editing with a strong commercial editor accumulates visible noise over 100 steps, and the NR-IQA metrics that an agent might use to detect this noise fail to do so (Tang et al., 2026). CoSTA* and FaSTA* show that cost deserves to be optimized explicitly. Modelling tool runtime in the search heuristic, and reusing mined subroutines, reduces cost without giving up the safety net of search (Gupta et al., 2025, 2026). A further trend is *editability as the output*. ReDesign and DrawAI recover layered, editable design structures from raster images, and MiLDEAgent edits multi-layer design documents. These systems reframe editing as the manipulation of a structured representation rather than of pixels (Yun et al., 2026; Cao et al., 2026; Lin et al., 2026b).

6.4. Closed-loop generation and workflow composition

Closed-loop generation (Table 7) is included only when visual evidence changes a later action. The lineage runs from iterative prompt refinement with an MLLM critic in Idea2Img (Yang et al., 2024b) and latent corrections controlled by an LLM in SLD (Wu et al., 2024d) to systems that orchestrate many generators and editors (Wang et al., 2024c; Chen et al., 2025a). Open-loop planners such as LayoutGPT, LLM-grounded diffusion, and RPG (Feng et al., 2023; Lian et al., 2024; Yang et al., 2024a) supplied the ideas for layout planning, but they do not observe their outputs and are therefore treated as foundations. Generation agents differ from editing agents mainly in what they observe. Instead of checking preservation, they check compositional correctness (objects, attributes, and relations),

aesthetics, and increasingly factual knowledge, for which retrieval-augmented agents search the web before generating (Bian et al., 2026; Zhang et al., 2026). MuLan generates one object at a time and regenerates it whenever a VLM finds that the prompt is violated (Li et al., 2024b), and CompAgent corrects attribute errors through verification and human feedback (Wang et al., 2024d). GenAgent calls image generators as tools and refines its outputs through judgment and reflection, with a policy trained by SFT and agentic reinforcement learning (Jiang et al., 2026a), and PosterCraft refines posters with suggestions from a fine-tuned VLM critic (Chen et al., 2025c). Multi-agent role structures are especially common in this family. Talk2Image routes hierarchical goals to analyst, generator, editor, and judge roles through a blackboard (Ma et al., 2026), and MCCD propagates evaluator feedback backward through specialized scene-parsing agents (Li et al., 2025b). Muses plans a 3D layout, retrieves or generates 3D assets, and renders control signals for 3D-controllable generation (Ding et al., 2025). Anywhere repaints only the foreground regions that its mask check finds corrupted (Tianyidan et al., 2025), and AuDiffusion swaps ControlNets or re-weights prompt tokens when its scores fall below a threshold (An et al., 2025). Correction after generation is becoming a task of its own. Agentic Retoucher localizes small distortions in limbs, faces, and text and repairs them with targeted inpainting (Shen et al., 2026b).

Workflow composition moves the agent one level up. Instead of calling tools, it writes the program that calls them. ComfyBench provided 200 tasks for evaluating agents that design ComfyUI workflows (Xue et al., 2025). Subsequent systems introduced multi-agent generation with reinforcement learning (Huang et al., 2025b), reasoning models trained on 4,000 workflows (Xu et al., 2026b), search guided by validation (Su et al., 2026), interactive assistants for workflow development (Xu et al., 2025), knowledge-centric fine-tuning (Li et al., 2026c), and self-evolving skill libraries (Li et al., 2026b). Workflow agents enjoy a property that pixel-level agents lack, since their outputs can be checked deterministically for validity and re-executed exactly. GenRouter addresses the opposite problem, a mismatch between the difficulty of a request and the compute spent on it, by routing each prompt to the cheapest adequate agentic workflow (Chen et al., 2026a).

6.5. Cross-task observations

Across tasks, four regularities emerge. (1) *The loop is the same but the judge differs.* Restoration uses IQA metrics and degradation-specific VLMs, retouching uses aesthetic and preference models, editing uses MLLM judges of instruction following and preservation, and generation uses judges of composition and text–image alignment. (2) *Interpretability follows the actions.* Parameters of professional software and programs yield auditable traces, whereas generative editors yield flexible but opaque transitions. (3) *Cost scales with exploration.* Search and reflection improve robustness at a multiplicative cost, which learned policies, reuse of experience, and routing try to reduce. (4) *Evidence is uneven.* Restoration agents are evaluated quantitatively on shared synthetic benchmarks, and retouching agents increasingly on dedicated benchmarks and user studies. Editing and generation agents are evaluated largely by MLLM judges, often from the same family of models that powers the agent.

7. Application domains

Application domains change what a good decision is. They impose constraints on fidelity (no hallucinated anatomy, preserved radiometry), downstream objectives (segmentation, detection, diagnosis), latency budgets, and requirements on provenance. Table 8 summarizes the domains covered by the reviewed systems.

Medical imaging Medical agents show both the promise and the stakes of the paradigm. SurgVisAgent identifies the category and severity of distortions in endoscopic images and selects among low-light enhancement, overexposure correction, motion-blur removal, and smoke removal, using a domain prior model and chain-of-thought reasoning in context (Lei et al., 2025). A denoising framework for positron emission tomography (PET) mirrors the workflow of clinical experts. VLM and LLM agents assess image quality and lesion status, choose denoising models and parameters, and roll back unsatisfactory results. On data acquired at 1/20 and 1/50 of the normal dose, it reported a higher PSNR and SSIM than UNet, GAN, and diffusion baselines (Yu et al., 2026a). AgentPolyp optimizes its enhancement policy for the segmentation of colon polyps, with a reward that weights the Dice overlap score at 0.9 and perceptual quality at 0.1 (Wang et al., 2025b). This is an explicit statement that "better" depends on the clinical task. At the research level, an agentic loop that edits, runs, and re-scores solvers for computed tomography (CT) reconstruction found that mild input noise nearly inverts a ranking obtained on idealized data (Maier et al., 2026). Agents tuned on clean benchmarks can therefore be confidently wrong in deployment. Medical use will require evidence beyond PSNR, including consistency of the result with the measured data, quantitative fidelity at the level of lesions, and traceable decisions.

Table 8
Application domains, representative systems, and the requirements they impose on the agent.

Domain	Representative systems	Main requirements
Medical imaging	SurgVisAgent (Lei et al., 2025), PET denoising agent (Yu et al., 2026a), AgentPolyp (Wang et al., 2025b), CT reconstruction loop (Maier et al., 2026)	No hallucinated structure, consistency with measurements, traceability
Remote sensing	EORestore-Agent (Qi et al., 2026), Multi-Weather DomainShifter (Wang et al., 2025c)	Radiometric fidelity, no reference at inference
Autonomous driving	JarvisIR (Lin et al., 2025a), CAMEO (Pu et al., 2026)	Downstream perception, latency
Photography and art	PhotoArtAgent, JarvisArt, PhotoAgent (Chen et al., 2025b; Lin et al., 2025b; Yao et al., 2026), CREA, EmoAgent (Venkatesh et al., 2025; Mao et al., 2026), sketching agents (Vinker et al., 2025; Qiu et al., 2024)	Subjective goals, personalization, reversible edits
Design and e-commerce	ReDesign, DrawAI, MiLDEAgent (Yun et al., 2026; Cao et al., 2026; Lin et al., 2026b), DiagramAgent (Wei et al., 2025), PosterCraft (Chen et al., 2025c), GMO-E^{2}DIT (Guo et al., 2026)	Editable layers, text rendering, product fidelity
Security and forensics	CraftAgent (Li et al., 2026a), RedEdit (Lin et al., 2026a), OrchJail (Chen et al., 2026b), VALOR (Zhao et al., 2026b)	Misuse resistance, safe tool use, provenance

Remote sensing EORestore-Agent is the clearest demonstration that agents built for natural images do not transfer automatically. On a synthetic Landsat-8 benchmark with six degradation types, it improved PSNR by 2.3–3.2 dB over the strongest retrained all-in-one baseline on composites of two to six degradations, whereas agents for natural images, applied without adaptation, fell below the degraded input in 17 of 18 settings. Its key design is a relative quality scorer trained with full-reference supervision, together with an acceptance rule that commits a step only when no predicted metric worsens and the predicted gain in PSNR is positive (Qi et al., 2026). Multi-Weather DomainShifter uses an LLM agent to transfer aerial images between weather conditions, so that downstream models can cope with the resulting domain shift (Wang et al., 2025c). Remote sensing thus makes the general lesson concrete. When no reference is available at inference, the agent's verifier must be calibrated for the domain.

Autonomous driving JarvisIR uses a VLM to control expert restoration models under combined adverse weather conditions. It is trained with supervised fine-tuning and with alignment from human feedback on unpaired real data, and it introduces CleanBench, which contains 150K synthetic and 80K real instruction–response entries. JarvisIR reports a 50% improvement in the average of perception metrics on CleanBench-Real (Lin et al., 2025a). CAMEO serves the opposite need and edits driving scenes under structural control in order to insert anomalies (Pu et al., 2026). Here the objective is not visual quality but downstream perception, which changes the appropriate reward and makes latency a hard constraint.

Photography, art, design, and commerce Creative domains drive the use of professional tools and explanations. PhotoArtAgent explains its artistic rationale and parameter choices. In user studies it was preferred over automated tools and produced results comparable to those of professional artists (Chen et al., 2025b). Sketching agents range from a dual-agent reinforcement-learning approach to realistic sketching (Qiu et al., 2024) to SketchAgent, which shares a canvas with the user (Vinker et al., 2025). Design and commerce require editability and auditability rather than only pixels. GMO-E^{2}DIT builds region-grounded editing agendas for e-commerce images, where partial or unrequested changes are commercially unacceptable (Guo et al., 2026). ReDesign evaluates editability by replaying 14,796 controlled edits on 909 reconstructed Figma files (Yun et al., 2026), and PosterCraft targets posters, where text rendering, layout, and aesthetics must be correct at the same time (Chen et al., 2025c).

Security and forensics Agentic editing has a dual use. The same closed loop that refines an edit can manufacture realistic manipulations. CraftAgent combines scene perception, planning of edits, execution, quality validation, and reflection to build a benchmark of 100K images for localizing manipulations (Li et al., 2026a). RedEdit formulates the evasion of image-safety classifiers as MCTS over sequences of edits and reports that fewer than two edits on average

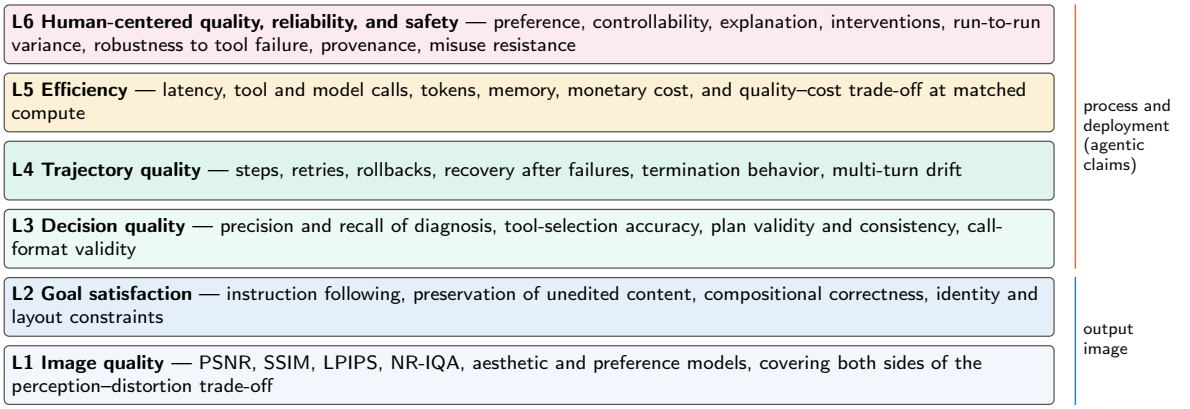


Figure 5: Six-layer evaluation framework. The lower layers concern the output image, and the upper layers concern the decision process and its deployment. Claims about agency require evidence from layers L3–L6, not only from L1 and L2.

allow 76.2% of unsafe images to evade detectors while retaining 93.0% of their malicious content (Lin et al., 2026a). OrchJail shows that tool-calling text-to-image agents can be jailbroken through orchestrations of individually benign steps (Chen et al., 2026b), while VALOR is a defensive agent that rewrites unsafe prompts and verifies the output before release (Zhao et al., 2026b). Safety evaluation must therefore consider whole trajectories, not only prompts or final images.

8. Evaluation of agentic image processing

An agentic paper typically claims that a *decision process* is better: it diagnoses more accurately, chooses better tools, recovers from failures, stops appropriately, or reaches the same quality at lower cost. Most evaluations, however, measure only the final image. We organize evaluation into six layers (Fig. 5), relate metrics to the claims they can support (Table 9), review benchmarks (Table 10), and propose a reporting checklist (Table 11).

8.1. L1: Image quality

Restoration agents report full-reference distortion metrics such as PSNR and SSIM and the perceptual metric LPIPS (Zhang et al., 2018) on synthetic benchmarks. On real images they report no-reference metrics such as MUSIQ, CLIP-IQA, and MANIQA (Ke et al., 2021; Wang et al., 2023a; Yang et al., 2022) and scores predicted by MLLMs, such as Q-Align (Wu et al., 2024c). Generation and editing agents report human-preference models such as ImageReward and HPSv2 (Xu et al., 2023; Wu et al., 2023b). Three cautions apply. First, metrics on opposite sides of the perception–distortion trade-off (Blau and Michaeli, 2018) can move in opposite directions under the same agent, as Restore-R1 documents when it adds more actions (Lu et al., 2026). Reporting only one side is therefore misleading. Second, metrics used as rewards or verifiers inside the loop should not be the main evaluation metric, although this practice is still common. Third, NR-IQA can fail exactly where agents create new artifacts: none of 21 NR-IQA metrics consistently ranked heavily degraded, 100-step iteratively edited images below clean ones (Tang et al., 2026). Systematic protocols for real-world super-resolution that report distributions rather than averages, such as SEAL (Zhang et al., 2024), and IQA datasets for perceptual restoration and generated images, such as PIPAL and AIGIQA-20K (Gu et al., 2020; Li et al., 2024a), provide better grounding.

8.2. L2: Goal satisfaction

For editing and generation, the question is whether the instruction was fulfilled without unwanted change. Benchmarks include MagicBrush for single- and multi-turn editing (Zhang et al., 2023a), EditVal (Basu et al., 2023), I2EBench (Ma et al., 2024), ImagenHub (Ku et al., 2024b), ImgEdit (Ye et al., 2025), and, for compositional generation, T2I-CompBench and GenEval (Huang et al., 2025a; Ghosh et al., 2023). Metrics range from feature similarity computed with CLIP or DINO and pixel distances to faithfulness measured by question answering (Hu et al., 2023) and MLLM judges that explain their scores (Ku et al., 2024a). EdiVal-Agent decomposes multi-turn editing evaluation into object-centric checks of instruction following, content consistency, and quality (Chen et al., 2025d). Preservation

deserves a separate measurement, because aggregated "overall" scores hide the trade-off between doing what was asked and leaving the rest intact.

8.3. L3: Decision quality

Decision quality asks whether individual choices were right, independently of how the image turned out. Measurable quantities include degradation-identification precision and recall (Zhu et al., 2025b), degradation-classification accuracy (Wei et al., 2026), tool-selection accuracy per request type (Li et al., 2025a), the validity rate of structured tool calls (Hang et al., 2025), workflow format validity and pass rates (Xu et al., 2026b; Huang et al., 2025b), and consistency of plans under re-ordering of the tool list (Zhu et al., 2025b). Such measures require ground truth for decisions—known degradation composition, oracle tool rankings, or executable specifications—which synthetic benchmarks can provide. They are reported by a minority of papers but are the most direct evidence for the claim that a controller "understands" the task.

8.4. L4: Trajectory quality

Trajectory metrics characterize the whole decision sequence: number of steps and tool calls, rollback and retry rates, recovery rate after a failed step, termination behavior under different budgets, and drift over turns. Exemplary analyses include MIRA's study of steps actually used under budgets from three to seven (Zeng et al., 2026), CCA's early-stopping analysis (Hang et al., 2025), and AgenticIR's ablations of reflection and rollback (Zhu et al., 2025b). Multi-turn editing benchmarks with long instruction chains, such as the LongBench-T2I-Edit benchmark introduced with RefineEdit-Agent (Liang et al., 2025), and iterative-degradation datasets (Tang et al., 2026) begin to make trajectory failure measurable. A recent review of restoration controllers also advocates process-aware evaluation (Yang et al., 2026).

8.5. L5: Efficiency

Efficiency is a central property of agentic systems, because exploration multiplies computation. Useful measures include wall-clock latency per image, number of tool and model calls, tokens consumed by language models, GPU memory, and monetary API cost, reported together with quality as a trade-off curve. Reported examples include runtime comparisons between search-based and learned restoration agents (Lu et al., 2026), TIR-Agent's 2.5× speed-up (Jia et al., 2026), EvoIR-Agent's 61% efficiency gain (Zhuang et al., 2026), HybridAgent's 0.08–0.13 s fast path against 0.75–1.05 s without it (Li et al., 2025a), MIRA's end-to-end latency of about 48 s per edit on an H100, dominated by calls to the diffusion editor rather than by reasoning (Zeng et al., 2026), and FlowTool's 50× lower latency than MLLM editing agents (Le et al., 2026). Comparisons should be made at matched compute. In language tasks, many reflection and debate strategies do not beat repeated sampling once every generated token is counted (Mirzaei, 2026), yet analogous best-of-N baselines are rarely run for image agents.

8.6. L6: Human-centered quality, reliability, and safety

The top layer asks whether people can use and trust the system: preference and satisfaction in user studies (Hang et al., 2025; Wei et al., 2026; Chen et al., 2025b), controllability and explanation quality, required interventions, robustness to perturbed inputs and tool failures, run-to-run variance, provenance of each edit, and resistance to misuse (Chen et al., 2026b; Lin et al., 2026a). Monitoring frameworks for deployed agentic systems argue for continuous, multi-dimensional evaluation rather than one-off benchmark scores (Shukla, 2026). Very few image-processing agents report variance across runs or seeds, although stochastic generators and sampled language-model outputs make trajectories non-deterministic.

8.7. A reporting checklist

Table 11 consolidates these observations into a checklist. It is deliberately modest, since each item is cheap compared with building the system, and its purpose is to make agentic claims falsifiable and comparable across papers.

9. Discussion and design guidelines

9.1. What does agentic control add?

The reviewed studies support four answers, each with caveats.

Composition beyond any single tool. When degradations or instructions combine, agents that sequence and verify specialist tools outperform both individual tools and random sequences on the benchmarks where they were developed

Table 9
Which evidence can support which claim about an agentic image-processing system.

Claim	Sufficient evidence	Insufficient on its own
"The output is better"	L1 and L2 metrics on held-out data, excluding the agent's internal judges, and a human study	Scores from the judge used inside the loop
"The agent diagnoses correctly"	Precision and recall of degradation or defect detection against the known composition (L3)	Final-image PSNR or MLLM score
"Planning matters"	Comparison with random and fixed orders of the same tools, and consistency of plans (L3)	Comparison with a single all-in-one model
"Reflection helps"	Ablation at matched compute and recovery rate after injected failures (L4)	Ablation that also removes compute
"The agent is efficient"	Latency, calls, tokens, and cost plotted against quality (L5)	Number of trainable parameters
"Memory helps"	Performance over a stream of tasks, transfer to new tools or degradations, and analysis of forgetting (L3–L5)	Single-episode results
"Multi-agent design helps"	Comparison with a single controller with identical tools and budget	Comparison with a weaker single model
"The system is reliable and safe"	Run-to-run variance, robustness to tool failures and adversarial requests, and an audit trail (L6)	Average benchmark score

(Zhu et al., 2025b; Jiang et al., 2026b; Wang et al., 2024c). The size of the gain depends on the tools and the benchmark, and agents do not escape the limits of their available actions (Li et al., 2025a; Kong et al., 2026).

Recovery from failure. Verification with rollback or alternative branches measurably improves results in ablations (Zhu et al., 2025b; Wang et al., 2024c), and conservative acceptance rules prevent harmful steps (Qi et al., 2026; Yao et al., 2026). Recovery is only as good as failure detection, however, and current judges miss important failure classes (Tang et al., 2026).

Interpretability and control. Agents that act through parameterized tools or programs produce inspectable traces, such as Lightroom settings, tool sequences, and workflow graphs, that support explanation, correction, and auditing (Lin et al., 2025b; Chen et al., 2025b; Guo et al., 2026; Xue et al., 2025). This is a distinctive advantage over monolithic generators for professional and regulated settings.

Extensibility. Registry cards, capability tables, and plug-in interfaces allow new tools to be added without retraining the controller (Jiang et al., 2026b; Gupta et al., 2025; Chen et al., 2026c), whereas trained controllers bound to a fixed tool pool require partial retraining (Choi and Oh, 2026). This creates a tension, because the methods that are most efficient at inference, namely learned policies, are the least extensible, and vice versa.

What agentic control does *not* yet demonstrate convincingly is superiority at matched compute. Few papers compare an agent with best-of-N sampling of the strongest single tool under the same budget, or a multi-agent design with a single controller using the same tools. Given evidence from language-model reasoning that reflection and debate are not reliably better than repeated sampling at equal token cost (Mirzaei, 2026), such comparisons are the most important missing experiment in the field.

9.2. An expert-systems reading

From the perspective of expert and intelligent systems, current image-processing agents re-implement the classic architecture with learned components. Tool cards, degradation-order priors, capability matrices, and causal memory graphs (Jiang et al., 2026b; Chen et al., 2026c; Gao et al., 2026) constitute a *knowledge base*. LLM planners and search procedures constitute an *inference engine*, MLLM and IQA judges play the role of *evaluation functions*, and experience pools and skill libraries (Zhuang et al., 2026; Shao et al., 2026; Gupta et al., 2026) implement *knowledge acquisition*. This reading suggests transferable lessons. Knowledge should be explicit, versioned, and validated, as in FaSTA*, which accepts a mined rule only when it yields a measured net benefit (Gupta et al., 2026). Inference should be explainable, and the evaluation function should be validated independently of the system it guides. It also highlights an unusual property. In image processing, the knowledge base contains not only symbolic facts but also learned operators whose behavior is distribution-dependent and partly unpredictable, so the knowledge base must be continuously re-validated by execution.

Table 10
Benchmarks and datasets used or introduced by agentic image-processing studies.

Benchmark	Task	Content and what it measures
MiO100 (Zhu et al., 2025b)	Restoration	Test groups with two or three mixed degradations, widely used for restoration agents (L1)
CleanBench (Lin et al., 2025a)	Restoration	150K synthetic and 80K real instruction–response entries for driving images in adverse weather (L1, L3)
Q-Degrade (Zhou et al., 2025)	Restoration	Ordered mixtures of eight degradation types for perception training (L3)
SEAL (Zhang et al., 2024)	Super-resolution	Systematic evaluation of real-world super-resolution over a space of degradations (L1)
MMArt (Lin et al., 2025b)	Retouching	Pairs of instructions and Lightroom configurations (L1–L3)
UGC-Edit (Yao et al., 2026)	Retouching	7,000 photos with an aesthetic reward model and a test set of 1,017 photos (L1, L2)
MagicBrush (Zhang et al., 2023a)	Editing	Manually annotated single- and multi-turn instruction editing (L2)
I2EBench, ImgEdit (Ma et al., 2024; Ye et al., 2025)	Editing	Multi-dimensional instruction-editing evaluation (L2)
EdiVal-Agent (Chen et al., 2025d)	Editing	Object-centric automated evaluation of multi-turn editing (L2, L4)
LongBench-T2I-Edit (Liang et al., 2025)	Editing	500 images with complex multi-turn instructions over nine visual dimensions (L2, L4)
EditFHF-15K (Xu et al., 2026a)	Editing	15K edited images with region-level failure annotations and mean opinion scores (L1, L2)
Banana100 (Tang et al., 2026)	Editing	28,000 images from 100-step iterative editing; tests NR-IQA failure (L1, L4)
T2I-CompBench, GenEval (Huang et al., 2025a; Ghosh et al., 2023)	Generation	Compositional attribute, relation, and object alignment (L2)
ComfyBench (Xue et al., 2025)	Workflow	200 ComfyUI tasks, 3,205 annotated nodes (L2, L3)
FlowBench (Huang et al., 2025b)	Workflow	13,571 pairs of workflows and descriptions, with format and pass-rate metrics (L3)
Figma Edit Replay (Yun et al., 2026)	Design	909 Figma files, 14,796 replayed edits measuring editability (L2)

Table 11
Reporting checklist for agentic image-processing studies.

Item	What to report
Tool library	Exact tools and versions, their individual performance, and how new tools are added
Controller	Model identity and version (including the release date for proprietary models), prompts, decoding settings, training data, and rewards
Judges	Which judges are used inside the loop, their calibration on the target domain, and evaluation with judges different from the in-loop ones
Baselines	Best single tool, all-in-one model, random and fixed orders over the same tools, best-of-N at matched compute, single-controller variant of multi-agent designs
Decisions	Diagnosis accuracy, tool-selection accuracy, plan consistency
Trajectories	Steps, calls, rollbacks, recovery after injected failures, termination statistics
Cost	Latency, tool and model calls, tokens, GPU memory, and monetary cost, plotted against quality
Variance	Results over seeds and repeated runs, and sensitivity to the order of the tool list
Real-world evidence	Evaluation on real, non-synthetic inputs and user studies with protocol details and participant counts
Traces	Release of full trajectories (actions, parameters, intermediate images) for audit and reuse

9.3. Choosing a design

Figure 6 condenses the evidence into a decision guide. We emphasize that it reflects tendencies in reported results, not controlled comparisons.

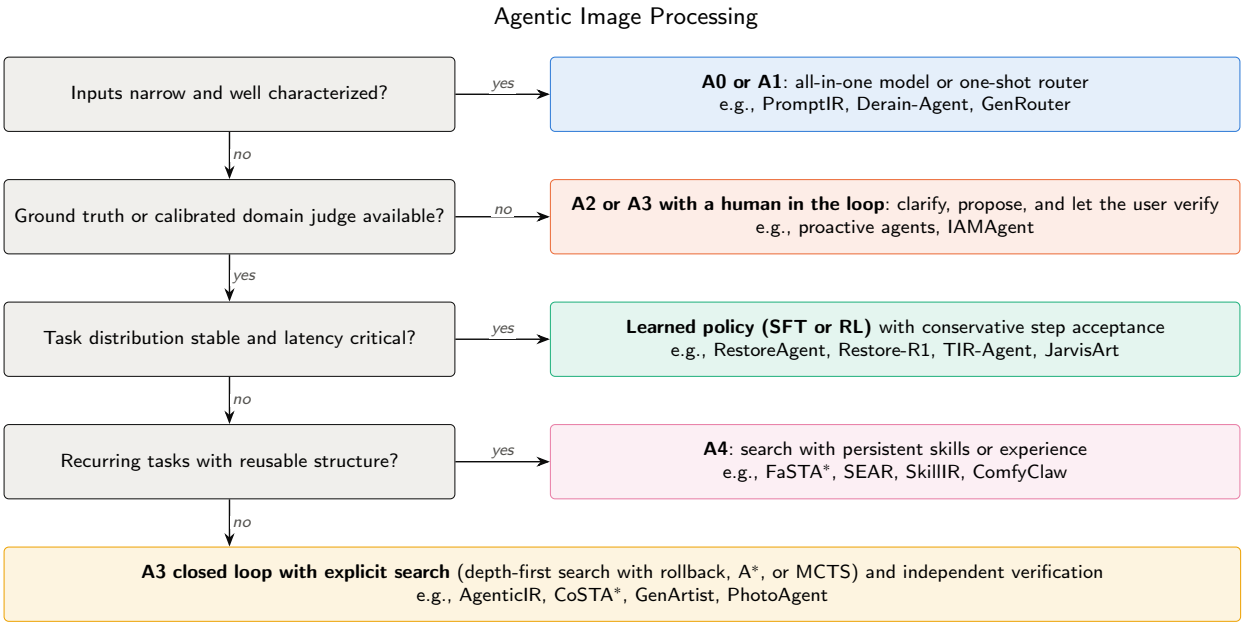


Figure 6: Decision guide for selecting an agentic design, derived from the reviewed evidence. Each leaf names representative systems.

G1. Start from the lowest autonomy that solves the task. If inputs are narrow and well characterized, for example a single family of degradations in a fixed domain, an all-in-one model or an A1 router is cheaper, deterministic, and easier to validate (Yu et al., 2026c; Chen et al., 2026a).

G2. Invest in the tools before the planner. Better tools, such as tools trained on mixed degradations, tools trained jointly, localized or layered editors, and foundation-model executors, often deliver larger gains than more elaborate planning (Li et al., 2025a; Zhu et al., 2026a; Ye et al., 2026; Kong et al., 2026).

G3. Calibrate the judge to the domain and keep it separate from the actor. Use domain-trained or full-reference-trained relative scorers where possible (Qi et al., 2026), ensembles of heterogeneous judges (Zhou et al., 2025; Cui et al., 2026), and evaluators anchored to human labels (Lin et al., 2025c). The judge used inside the loop should never provide the main reported result.

G4. Prefer conservative acceptance and explicit stopping. Commit a step only when an improvement is predicted, keep the best state found so far, and include a stop action. These rules turn verification errors into lost opportunities rather than damaged images (Qi et al., 2026; Yao et al., 2026; Yu et al., 2018).

G5. Learn a policy when the input distribution is stable, and search when it is not. Offline search distilled into SFT and RL policies reduces latency (Chen et al., 2024; Lu et al., 2026; Jia et al., 2026), and dual-process designs keep slow search as a fallback for novel inputs (Gupta et al., 2026; Cui et al., 2026; Li et al., 2025a).

G6. Make memory explicit, conditional, and revisable. Store applicability conditions and failure cases, not only successes (Shao et al., 2026), and support updating and forgetting (Gao et al., 2026).

G7. Use multiple agents only for a reason. Role separation is justified when roles need different models, information, or incentives, as an independent critic does. Otherwise, a single controller with the same tools is a necessary baseline.

G8. Prefer reversible, parameterized, or programmatic actions when auditability matters. Professional-tool parameters and workflow programs yield traces that experts can review and that can be re-executed exactly (Lin et al., 2025b; Li et al., 2026b).

9.4. Failure chains and mitigations

Failures in agentic image processing are rarely isolated. They propagate along the loop. Table 12 lists the most frequent chains that we identified in the experiments and limitation analyses of the reviewed papers, together with the mitigations those papers propose.

Table 12
Common failure chains and the mitigations proposed in the reviewed systems.

Failure chain	Evidence	Mitigations
Misdiagnosis → wrong tool → accepted by a judge with the same blind spot	Low perceiver recall for blur types (Zhu et al., 2025b), degradations hallucinated by an MLLM (Li et al., 2025a)	One CoT question per degradation (Zhou et al., 2025), majority voting, reassessment of the remaining degradations (Shao et al., 2026)
Judge exploitation (reward hacking)	Rising no-reference scores with falling fidelity (Lu et al., 2026), NR-IQA blind to accumulated artifacts (Tang et al., 2026)	Adaptive rewards combining several metrics (Jia et al., 2026), tournaments among MLLM judges (Cui et al., 2026), scorers trained with full references (Qi et al., 2026)
Unrequested changes in editing → drift over turns	Error accumulation in multi-turn editing (Shen et al., 2026a; Hang et al., 2025)	Execution on layers or crops (Ye et al., 2026), sub-programs conditioned on masks (Guo et al., 2026), recall of the original input (Qiu et al., 2026)
Generative tool hallucination	Shadows misread as geometry or material (Hu et al., 2026)	Candidate sampling with physics-grounded selection (Hu et al., 2026)
Combinatorial search cost	Runtime growing with degradation count (Lu et al., 2026)	Order priors (Jiang et al., 2026b), A* search with cost heuristics (Gupta et al., 2025), learned policies, reuse of experience (Zhuang et al., 2026)
Tool-call format and interface errors	Format errors depend strongly on the LLM (Hang et al., 2025)	Two-level tool manuals (Hang et al., 2025), typed interfaces and automatic reverts (Li et al., 2026b)
Domain shift of controller and judge	Agents for natural images below the input PSNR in remote sensing (Qi et al., 2026)	Domain-specific scorers and tool pools (Qi et al., 2026; Lei et al., 2025)
Unsafe orchestration	Benign steps composing harmful outputs (Chen et al., 2026b)	Output verification and staged rewriting (Zhao et al., 2026b)

9.5. Quality, cost, and autonomy

Autonomy is not free. Each additional level adds computation for search and verification, new points of failure such as judges and memory, and variance from stochastic planners and generators. At the same time, higher autonomy allows the system to cover a broader input distribution with fewer human interventions. The appropriate operating point depends on the deployment. An interactive photo editor needs low latency and predictable behavior, an offline service that restores archives can afford search, and a clinical pipeline needs traceability above all. Reporting the trade-off between quality and cost, together with the number of human interventions (Section 8), would let practitioners choose with evidence rather than by analogy.

10. Open challenges and research directions

1. Trustworthy judges and rewards Every closed-loop and RL-trained system depends on a judge, and the evidence of judge failure is mounting. NR-IQA metrics fail to detect accumulated editing artifacts (Tang et al., 2026), perceptual rewards trade away fidelity (Lu et al., 2026), natural-image quality signals mislead in other domains (Qi et al., 2026), and self-refinement can exploit shared evaluator weaknesses (Pan et al., 2024; Xu et al., 2024). The field needs judges that are calibrated for each domain, aware of their own uncertainty, sensitive to the artifacts introduced by the tools they supervise, and tested adversarially. It also needs training procedures that limit over-optimization, for example adaptive rewards that combine several metrics (Jia et al., 2026), tournaments among judges (Cui et al., 2026), or evaluators anchored to human labels (Lin et al., 2025c). Verification strategies developed for visual reasoning agents may transfer here. CaptchaAgent, for instance, re-examines ambiguous regions at higher resolution and accepts a decision only if it remains stable under transformations that should not change the answer (Sun et al., 2026a). Analogous consistency tests could expose judges that reward artifacts in edited or restored images.

2. State representation and credit assignment over long horizons Most agents represent state as the current image plus a textual summary. Long multi-turn edits and multi-step restorations need representations that track what has been changed, what must be preserved, and which residual problems remain—layered scene graphs (Yu et al., 2026b), state graphs (Ye et al., 2026), or residual-degradation states (Shao et al., 2026). Assigning credit to individual actions when only the final image is judged remains an open problem. Rewards for intermediate steps and counterfactual replay of trajectories are promising directions.

3. Tool interoperability and capability modelling Tool descriptions are hand-written, quickly outdated, and incomparable across systems. Standard interfaces, such as MCP-style tool servers (Ye et al., 2026), typed workflow graphs (Li et al., 2026b), and empirically updated capability profiles (Chen et al., 2026c), point toward tool ecosystems in which capabilities are measured rather than described. Shared tool registries with benchmarked operating ranges would also make agent comparisons fairer.

4. Memory that generalizes and forgets Persistent memory appears in only a few systems. Open questions include transfer of experience across tool versions and domains, detection of stale or contaminated entries, privacy of stored user images and preferences, and the risk that reusing successful trajectories homogenizes outputs. Causal and conditional representations (Gao et al., 2026; Shao et al., 2026) are early steps.

5. Efficiency and adaptive test-time computation Search, verification, and large controllers make agents slow and expensive. Promising directions are routing inputs to the cheapest adequate workflow (Chen et al., 2026a), dual-process control with a fast and a slow path (Li et al., 2025a; Gupta et al., 2026), small specialized controllers (Lu et al., 2026; Huang et al., 2026), non-autoregressive parameter generation (Le et al., 2026), and learned stopping. Typed decision models, also called System One models, offer another route. Models such as Jev return calibrated probabilities over a fixed set of options in a single forward pass instead of generating text token by token (Almeida, 2026). Early studies apply them to forced-choice questions about images (Yu and Yao, 2026) and to zero-shot prediction of video quality (Robitza, 2026). To our knowledge, no image-processing agent yet uses such a model to choose tools, accept or reject a step, or decide when to stop, although these are exactly the fast and frequent decisions that dominate the cost of an agentic loop. Evaluations at matched compute should accompany every efficiency claim.

6. Real-world generalization and benchmarks Restoration agents are largely evaluated on synthetic mixtures of degradations, and editing and generation agents on benchmarks scored by MLLM judges. The field needs real-degradation benchmarks with decision-level annotations, multi-turn editing benchmarks with preservation ground truth, and process benchmarks that release trajectories. Ranking inversions under realistic noise (Maier et al., 2026) illustrate why.

7. Human–agent co-processing Professional users want control, explanation, and the ability to intervene. Research is needed on when an agent should ask rather than act (Hahn et al., 2025), how to explain image decisions faithfully, how to learn individual preferences without over-fitting (Chang et al., 2026), and how to measure workload and trust.

8. Safety, provenance, and accountability Agents that compose benign tools can produce harmful results (Chen et al., 2026b), evade detectors (Lin et al., 2026a), and generate realistic manipulations at scale (Li et al., 2026a). Trajectory-level safety checks, signed provenance records for every edit, and audit logs of tool calls should be part of the system design, particularly in medical, forensic, and journalistic settings.

9. Unified models versus orchestrated tools Unified understanding-and-generation models and strong instruction editors (Deng et al., 2025; Black Forest Labs et al., 2025; Wu et al., 2025) increasingly perform tasks that previously required tool chains, and HarnessIR shows that a foundation model can serve as the executor of an agent (Kong et al., 2026). Whether future systems internalize agency into one model, as Chain-of-Restoration and ImAgent partially do (Cao et al., 2024; Wang et al., 2025a), or keep explicit orchestration for auditability and extensibility is an open, empirical question that should be studied under matched compute and transparent evaluation.

10. Reproducibility Dependence on proprietary, versioned APIs, unreported prompts, and stochastic tools makes many results hard to reproduce. Open controllers, released trajectories, fixed tool versions, and the reporting checklist in Table 11 would make agentic image processing a cumulative science.

11. Conclusion

Image processing is moving from single mappings to decision processes over image states. This review defined agentic image processing by its behavior, as processing in which an identifiable controller uses observations of the input or of intermediate images to choose later actions. With this definition we analyzed representative systems for restoration, enhancement and retouching, editing, closed-loop generation, and workflow composition. A four-axis

taxonomy separated what a system aims to produce, how it decides, how adaptive it is, and how it is organized, and an autonomy scale from A0 to A4 made claims about autonomy checkable.

The synthesis yields a consistent picture. Across tasks, systems converge on a perceive–plan–act–verify–remember loop built from MLLM diagnosis, structured tool knowledge, explicit or learned search, judge-based verification, and, increasingly, persistent experience. Controllers are shifting from training-free orchestration of proprietary models toward supervised and reinforcement-learned policies, which makes reward and judge design the central technical problem. Agentic control demonstrably helps with composition, recovery, interpretability, and extensibility, but its advantages at matched compute and on real-world inputs remain under-documented, and the judges that close the loop are its weakest link. We therefore proposed a six-layer evaluation framework, a map from metrics to claims, a reporting checklist, and design guidelines intended to make future agentic claims comparable and falsifiable. If the community adopts process-aware evaluation as readily as it has adopted agent architectures, agentic image processing can mature from impressive demonstrations into reliable expert systems for imaging.

Data availability

The list of reviewed systems, with their characterization along the dimensions of the taxonomy, is provided as supplementary material.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used AI coding assistants (OpenAI Codex and Anthropic Claude) to help organize the literature and retrieve bibliographic metadata. After using these tools, the author wrote, reviewed and edited the content, checked bibliographic records and claims against the original papers, and takes full responsibility for the content of the published article.

References

Almeida, D., 2026. Introducing system one models & Jev. TypeSafe AI blog, https://typesafe.ai/blog/introducing-system-one-models-and-jev.

An, D., Zhang, W., Zhang, S., Lu, J., 2025. AuDiffusion: multi-agent controlled text-to-image generation with attention-enhanced mamba blocks. Complex & Intelligent Systems 12. doi:10.1007/s40747-025-02211-1.

Bai, S., Chen, K., Liu, X., Wang, J., Ge, W., Song, S., Dang, K., Wang, P., Wang, S., Tang, J., Zhong, H., Zhu, Y., et al., 2025. Qwen2.5-VL technical report. arXiv preprint arXiv:2502.13923 .

Basu, S., Saberi, M., Bhardwaj, S., Chegini, A.M., Massiceti, D., Sanjabi, M., Hu, S.X., Feizi, S., 2023. EditVal: benchmarking diffusion based text-guided image editing methods. arXiv preprint arXiv:2310.02426 .

Bian, F., Zheng, Z., Deng, W., Zhou, D., Luan, J., 2026. RS-Gen: a multi-stage agentic framework for reasoning and search-augmented image generation. arXiv preprint arXiv:2606.23221 .

Black Forest Labs, Batifol, S., Blattmann, A., Boesel, F., Consul, S., Diagne, C., Dockhorn, T., English, J., English, Z., Esser, P., Kulal, S., Lacey, K., et al., 2025. FLUX.1 kontext: Flow matching for in-context image generation and editing in latent space. arXiv preprint arXiv:2506.15742 .

Blau, Y., Michaeli, T., 2018. The perception-distortion tradeoff, in: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, IEEE. pp. 6228–6237. doi:10.1109/CVPR.2018.00652.

Brooks, T., Holynski, A., Efros, A.A., 2023. InstructPix2Pix: learning to follow image editing instructions, in: 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE. pp. 18392–18402. doi:10.1109/CVPR52729.2023.01764.

Cao, J., Meng, D., Cao, X., 2024. Chain-of-restoration: Multi-task image restoration models are zero-shot step-by-step universal image restorers. arXiv preprint arXiv:2410.08688 .

Cao, P., Kong, Q., Yin, X., Zhao, X., Yan, R., Song, Q., Zhang, Y., Yang, L., 2026. DrawAI: agentic benchmark and workflow for making raster images editable. arXiv preprint arXiv:2608.00548 .

Chang, Z., Duan, Z.P., Zhang, J., Guo, C.L., Liu, S., Chun, H., Park, H., Liu, Z., Li, C., 2026. PerTouch: VLM-Driven agent for personalized and semantic image retouching. Proceedings of the AAAI Conference on Artificial Intelligence 40, 2752–2759. doi:10.1609/aaai.v40i4.37264.

Chen, C.Y., Shi, M., Zhang, G., Shi, H., 2025a. T2I-Copilot: a training-free multi-agent text-to-image system for enhanced prompt interpretation and interactive generation, in: 2025 IEEE/CVF International Conference on Computer Vision (ICCV), IEEE. pp. 19396–19405. doi:10.1109/ICCV51701.2025.01803.

Chen, H., Li, W., Gu, J., Ren, J., Chen, S., Ye, T., Pei, R., Zhou, K., Song, F., Zhu, L., 2024. RestoreAgent: autonomous image restoration agent via multimodal large language models, in: Advances in Neural Information Processing Systems.

Chen, H., Tao, K., Wang, Y., Wang, X., Zhu, L., Gu, J., 2025b. PhotoArtAgent: intelligent photo retouching with language model-based artist agents. arXiv preprint arXiv:2505.23130 .

Chen, H.H., Hou, Z., Shu, W.J., Ruan, W., Xu, Y., Guo, L., Chen, Y.C., 2026a. GenRouter: unified workflow routing for agentic image generation. arXiv preprint arXiv:2608.16721 .

Chen, J., Wang, Y., Wang, J., Liu, Z., Wang, Q., Xu, F., 2026b. OrchJail: jailbreaking tool-calling text-to-image agents by orchestration-guided fuzzing, in: International Conference on Machine Learning.

Chen, S., Lai, J., Gao, J., Ye, T., Chen, H., Shi, H., Shao, S., Lin, Y., Fei, S., Xing, Z., Jin, Y., Luo, J., et al., 2025c. PosterCraft: rethinking high-quality aesthetic poster generation in a unified framework. arXiv preprint arXiv:2506.10741 .

Chen, T., Zhang, Y., Zhang, Z., Yu, P., Wang, S., Wang, Z., Lin, K., Wang, X., Yang, Z., Li, L., Lin, C.C., Xie, J., et al., 2025d. EdiVal-Agent: an object-centric framework for automated, fine-grained evaluation of multi-turn editing. arXiv preprint arXiv:2509.13399 .

Chen, Z., Zhang, Z., Zhang, C., Xu, Y., Yang, L., Liu, J., Li, K., Song, Y.Z., 2026c. PerfGuard: a performance-aware agent for visual content generation. arXiv preprint arXiv:2601.22571 .

Choi, S., Oh, J., 2026. DiTTo: scalable order-aware all-in-one image restoration agent. arXiv preprint arXiv:2605.30915 .

Conde, M.V., Geigle, G., Timofte, R., 2024. InstructIR: high-quality image restoration following human instructions, in: Computer Vision – ECCV 2024, Springer Nature Switzerland. pp. 1–21. doi:10.1007/978-3-031-72764-1_1.

Cui, S., Ji, F., Sun, G., Guo, Y., Tang, X., Li, J., Xu, F., 2026. Self-evolving agentic image restoration via deliberate planning and intuitive execution, in: Computer Vision – ECCV 2026, Springer Nature Switzerland. pp. 392–411. doi:10.1007/978-3-032-37422-6_22.

Deng, C., Zhu, D., Li, K., Gou, C., Li, F., Wang, Z., Zhong, S., Yu, W., Nie, X., Song, Z., Shi, G., Fan, H., 2025. Emerging properties in unified multimodal pretraining. arXiv preprint arXiv:2505.14683 .

Ding, Y., Zhuang, S., Li, K., Yue, Z., Qiao, Y., Wang, Y., 2025. Muses: 3D-Controllable image generation via multi-modal agent collaboration. Proceedings of the AAAI Conference on Artificial Intelligence 39, 2753–2761. doi:10.1609/aaai.v39i3.32280.

Fan, H., Wang, L., Chen, H., Huang, Z., Wu, J., Sheng, L., 2025. Multi-agent amodal completion: Direct synthesis with fine-grained semantic guidance, in: Proceedings of the 33rd ACM International Conference on Multimedia, ACM. pp. 9911–9919. doi:10.1145/3746027.3755225.

Feng, W., Zhu, W., Fu, T.j., Jampani, V., Akula, A., He, X., Basu, S., Wang, X.E., Wang, W.Y., 2023. LayoutGPT: compositional visual planning and generation with large language models, in: Advances in Neural Information Processing Systems.

Franklin, S., Graesser, A., 1997. Is it an agent, or just a program?: A taxonomy for autonomous agents, in: Intelligent Agents III Agent Theories, Architectures, and Languages, Springer Berlin Heidelberg. pp. 21–35. doi:10.1007/BFb0013570.

Fu, T.J., Hu, W., Du, X., Wang, W.Y., Yang, Y., Gan, Z., 2024. Guiding instruction-based image editing via multimodal large language models, in: International Conference on Learning Representations.

Furuta, R., Inoue, N., Yamasaki, T., 2019. Fully convolutional network with multi-step reinforcement learning for image processing. Proceedings of the AAAI Conference on Artificial Intelligence 33, 3598–3605. doi:10.1609/AAAI.V33I01.33013598.

Furuta, R., Inoue, N., Yamasaki, T., 2020. PixelRL: fully convolutional network with reinforcement learning for image processing. IEEE Transactions on Multimedia 22, 1704–1719. doi:10.1109/TMM.2019.2960636.

Gao, H., Chen, Y., Ma, L., 2026. Causal-AgentIR: self-evolving causal memory for adaptive image restoration agents. arXiv preprint arXiv:2607.21125 .

Ghosh, D., Hajishirzi, H., Schmidt, L., 2023. GenEval: an object-focused framework for evaluating text-to-image alignment, in: Advances in Neural Information Processing Systems.

Gu, J., 2025. Position: Agentic systems constitute a key component of next-generation intelligent image processing. arXiv preprint arXiv:2505.16007 .

Gu, J., Cai, H., Chen, H., Ye, X., Ren, J., Dong, C., 2020. Image quality assessment for perceptual image restoration: A new dataset, benchmark and metric. arXiv preprint arXiv:2011.15002 .

Guo, T., Chen, X., Wang, Y., Chang, R., Pei, S., Chawla, N.V., Wiest, O., Zhang, X., 2024. Large language model based multi-agents: A survey of progress and challenges, in: International Joint Conference on Artificial Intelligence, pp. 8048–8057.

Guo, Z., Liu, X., Ma, L., Wang, C., He, Y., Fu, X., Fu, J., Shan, X., Guo, S., Liu, L., Huang, J., Li, Y., 2026. GMO-E^{2}DIT: grounded multi-operation editing for e-commerce images. arXiv preprint arXiv:2607.00920 .

Gupta, A., Raj, R., Nguyen, D., Zhou, T., 2026. FaSTA*: fast-slow toolpath agent with subroutine mining for efficient multi-turn image editing, in: International Conference on Learning Representations.

Gupta, A., Velaga, N., Nguyen, D., Zhou, T., 2025. CoSTA*: cost-sensitive toolpath agent for multi-turn image editing. arXiv preprint arXiv:2503.10613 .

Gupta, T., Kembhavi, A., 2023. Visual programming: Compositional visual reasoning without training, in: 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE. pp. 14953–14962. doi:10.1109/CVPR52729.2023.01436.

Hahn, M., Zeng, W., Kannen, N., Galt, R., Badola, K., Kim, B., Wang, Z., 2025. Proactive agents for multi-turn text-to-image generation under uncertainty, in: International Conference on Machine Learning.

Hang, T., Gu, S., Chen, D., Geng, X., Guo, B., 2025. CCA: collaborative competitive agents for image editing. Frontiers of Computer Science 19. doi:10.1007/s11704-025-41244-0.

Hertz, A., Mokady, R., Tenenbaum, J., Aberman, K., Pritch, Y., Cohen-Or, D., 2023. Prompt-to-prompt image editing with cross attention control, in: International Conference on Learning Representations.

Hu, S., Xu, J., Samaras, D., Le, H., 2026. Domain-grounded candidate selection for agentic image editing: A shadow removal case. arXiv preprint arXiv:2608.06075 .

Hu, Y., He, H., Xu, C., Wang, B., Lin, S., 2018. Exposure: A white-box photo post-processing framework. ACM Transactions on Graphics 37, 26:1–26:17. doi:10.1145/3181974.

Hu, Y., Liu, B., Kasai, J., Wang, Y., Ostendorf, M., Krishna, R., Smith, N.A., 2023. TIFA: accurate and interpretable text-to-image faithfulness evaluation with question answering, in: 2023 IEEE/CVF International Conference on Computer Vision (ICCV), IEEE. pp. 20349–20360. doi:10.1109/ICCV51070.2023.01866.

Hu, Y., Liu, S., Tan, Z., Yang, X., Wang, X., 2025. Image editing as programs with diffusion models, in: Advances in Neural Information Processing Systems.

Huang, K., Duan, C., Sun, K., Xie, E., Li, Z., Liu, X., 2025a. T2I-CompBench++: an enhanced and comprehensive benchmark for compositional text-to-image generation. IEEE Transactions on Pattern Analysis and Machine Intelligence 47, 3563–3579. doi:10.1109/TPAMI.2025.3531907.

Huang, O., Ma, Y., Zhao, Z., Wu, M., Ji, J., Zhang, R., Hu, Z., Sun, X., Ji, R., 2025b. ComfyGPT: a self-optimizing multi-agent system for comprehensive ComfyUI workflow generation. arXiv preprint arXiv:2503.17671 .

Huang, Y., Huang, J., Liu, Y., Yan, M., Lv, J., Liu, J., Xiong, W., Zhang, H., Cao, L., Chen, S., 2025c. Diffusion model-based image editing: A survey. IEEE Transactions on Pattern Analysis and Machine Intelligence 47, 4409–4437. doi:10.1109/TPAMI.2025.3541625.

Huang, Y., Xie, L., Wang, X., Yuan, Z., Cun, X., Ge, Y., Zhou, J., Dong, C., Huang, R., Zhang, R., Shan, Y., 2024. SmartEdit: exploring complex instruction-based image editing with multimodal large language models, in: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE. pp. 8362–8371. doi:10.1109/CVPR52733.2024.00799.

Huang, Z., Wu, J.Z., Wang, Z., Cao, T., Chen, J., Fidler, S., Ling, H., Ren, X., 2026. APE: agentic prompt enhancer for image generation and editing. arXiv preprint arXiv:2606.00204 .

Jia, G., Zhang, Y., Hu, H., Zhao, S., Zhao, K., Sun, L., Long, X., Tian, K., Jiang, C., Liu, Z., Wang, K., Lian, S., et al., 2026. TIR-Agent: training an explorative and efficient agent for image restoration, in: Computer Vision – ECCV 2026, Springer Nature Switzerland. pp. 523–540. doi:10.1007/978-3-032-37281-9_32.

Jiang, J., Zuo, Z., Wu, G., Jiang, K., Liu, X., 2025. A survey on all-in-one image restoration: Taxonomy, evaluation and future trends. IEEE Transactions on Pattern Analysis and Machine Intelligence 47, 11892–11911. doi:10.1109/TPAMI.2025.3598132.

Jiang, K., Wang, Y., Zhou, J., Li, P., Liu, Z., Xie, C.W., Chen, Z., Zheng, Y., Zhang, W., 2026a. GenAgent: scaling text-to-image generation via agentic multimodal reasoning. arXiv preprint arXiv:2601.18543 .

Jiang, X., Li, G., Chen, B., Zhang, J., 2026b. Multi-agent image restoration. International Journal of Computer Vision 134. doi:10.1007/s11263-026-02792-5.

Jiang, Y., Zhang, Z., Xue, T., Gu, J., 2024. AutoDIR: automatic all-in-one image restoration with latent diffusion, in: Computer Vision – ECCV 2024, Springer Nature Switzerland. pp. 340–359. doi:10.1007/978-3-031-73661-2_19.

Kawar, B., Elad, M., Ermon, S., Song, J., 2022. Denoising diffusion restoration models, in: Advances in Neural Information Processing Systems 35, Neural Information Processing Systems Foundation, Inc. (NeurIPS). pp. 23593–23606. doi:10.52202/068431-1714.

Ke, J., Wang, Q., Wang, Y., Milanfar, P., Yang, F., 2021. MUSIQ: multi-scale image quality transformer, in: 2021 IEEE/CVF International Conference on Computer Vision (ICCV), IEEE. pp. 5128–5137. doi:10.1109/ICCV48922.2021.00510.

Kim, H., Kim, D.S., Yoon, S., Dernoncourt, F., Bui, T., Bansal, M., 2022. CAISE: conversational agent for image search and editing. Proceedings of the AAAI Conference on Artificial Intelligence 36, 10903–10911. doi:10.1609/aaai.v36i10.21337.

Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A.C., Lo, W.Y., Dollár, P., Girshick, R., 2023. Segment anything, in: 2023 IEEE/CVF International Conference on Computer Vision (ICCV), IEEE. pp. 3992–4003. doi:10.1109/ICCV51070.2023.00371.

Kong, X., Liu, S., Wu, R., Sun, L., Zhang, Z., Zhao, J., Wu, Y., Zhang, L., 2026. HarnessIR: harnessing multimodal foundation models for universal real-world image restoration. arXiv preprint arXiv:2610.10133 .

Kosugi, S., Yamasaki, T., 2020. Unpaired image enhancement featuring reinforcement-learning-controlled image editing software. Proceedings of the AAAI Conference on Artificial Intelligence 34, 11296–11303. doi:10.1609/AAAI.V34I07.6790.

Ku, M., Jiang, D., Wei, C., Yue, X., Chen, W., 2024a. VIEScore: towards explainable metrics for conditional image synthesis evaluation, in: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics. pp. 12268–12290. doi:10.18653/v1/2024.acl-long.663.

Ku, M., Li, T., Zhang, K., Lu, Y., Fu, X., Zhuang, W., Chen, W., 2024b. ImagenHub: standardizing the evaluation of conditional image generation models, in: International Conference on Learning Representations.

Le, T.L.V., Walton, S., Yoon, S., Kveton, B., Bui, T., Yang, E., Lai, V., 2026. FlowTool: controlling tool parameter in image retouching via flow matching. arXiv preprint arXiv:2609.35673 .

Lei, Z., Yu, H., Wu, J., Chen, Z., 2025. SurgVisAgent: multimodal agentic model for versatile surgical visual enhancement. arXiv preprint arXiv:2507.02252 .

Li, B., Li, X., Lu, Y., Chen, Z., 2025a. Hybrid agents for image restoration. arXiv preprint arXiv:2503.10120 .

Li, B., Liu, X., Hu, P., Wu, Z., Lv, J., Peng, X., 2022. All-in-one image restoration for unknown corruption, in: 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE. pp. 17431–17441. doi:10.1109/CVPR52688.2022.01693.

Li, C., Kou, T., Gao, Y., Cao, Y., Sun, W., Zhang, Z., Zhou, Y., Zhang, Z., Zhang, W., Wu, H., Liu, X., Min, X., et al., 2024a. AIGIQA-20K: a large database for AI-Generated image quality assessment, in: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), IEEE. pp. 6327–6336. doi:10.1109/CVPRW63382.2024.00636.

Li, L., Liang, D., Gao, Y., Huang, S.J., Chen, S., 2023. ALL-E: aesthetics-guided low-light image enhancement, in: Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence, International Joint Conferences on Artificial Intelligence Organization. pp. 1062–1070. doi:10.24963/ijcai.2023/118.

Li, M., Hou, X., Liu, Z., Yang, D., Qian, Z., Chen, J., Wei, J., Jiang, Y., Xu, Q., Zhang, L., 2025b. MCCD: multi-agent collaboration-based compositional diffusion for complex text-to-image generation, in: 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE. pp. 13263–13272. doi:10.1109/CVPR52734.2025.01238.

Li, S., Wang, R., Hsieh, C.J., Cheng, M., Zhou, T., 2024b. MuLan: Multimodal-LLM agent for progressive and interactive multi-object diffusion. arXiv preprint arXiv:2402.12741 .

Li, Z., Hu, Y., Wang, Q., Du, W., Jiang, H., Wu, J., Jiang, X., Han, J., 2026a. Impostor: An agent-curated benchmark for realistic AIGC manipulation localization. arXiv preprint arXiv:2606.04545 .

Li, Z., Liu, D., Liu, F., Zhou, Y., Wu, X., Chen, J., Xie, J., Wu, X., Sun, L., 2026b. COMFYCLAW: self-evolving skill harnesses for image generation workflows. arXiv preprint arXiv:2607.01709 .

Li, Z., Sun, L., Ming, R., Zhang, H., Paudel, D.P., Van Gool, L., Gu, J., 2026c. Knowledge-centric agents for workflow generation in ComfyUI, in: Computer Vision – ECCV 2026, Springer Nature Switzerland. pp. 306–323. doi:10.1007/978-3-032-37232-1_17.

Lian, L., Li, B., Yala, A., Darrell, T., 2024. LLM-grounded diffusion: Enhancing prompt understanding of text-to-image diffusion models with large language models. Transactions on Machine Learning Research .

Liang, J., Cao, J., Sun, G., Zhang, K., Van Gool, L., Timofte, R., 2021. SwinIR: image restoration using swin transformer, in: 2021 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW), IEEE. pp. 1833–1844. doi:10.1109/ICCVW54120.2021.00210.

Liang, Z., Sun, J., Ma, H., 2025. An LLM-LVLM driven agent for iterative and fine-grained image editing. arXiv preprint arXiv:2508.17435 .

Lin, W., Lin, Z., Zhou, Z., Li, J., Zhang, T., Xiong, H., Liu, L., 2026a. RedEdit: agentic red-teaming of image safety classifiers via MCTS-Guided photo-editing. arXiv preprint arXiv:2606.06140 .

Lin, X., He, J., Chen, Z., Lyu, Z., Dai, B., Yu, F., Ouyang, W., Qiao, Y., Dong, C., 2024. DiffBIR: towards blind image restoration with generative diffusion prior, in: European Conference on Computer Vision.

Lin, Y., Lin, Z., Chen, H., Pan, P., Li, C., Chen, S., Wen, K., Jin, Y., Li, W., Ding, X., 2025a. JarvisIR: elevating autonomous driving perception with intelligent image restoration, in: 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE. pp. 22369–22380. doi:10.1109/CVPR52734.2025.02084.

Lin, Y., Lin, Z., Lin, K., Bai, J., Pan, P., Li, C., Chen, H., Wang, Z., Ding, X., Li, W., Yan, S., 2025b. JarvisArt: liberating human artistic creativity via an intelligent photo retouching agent, in: Advances in Neural Information Processing Systems.

Lin, Y., Wang, L., Lin, K., Lin, Z., Gong, K., Li, W., Lin, B., Li, Z., Zhang, S., Peng, Y., Dai, W., Ding, X., et al., 2025c. JarvisEvo: towards a self-evolving photo editing agent with synergistic editor-evaluator optimization. arXiv preprint arXiv:2511.23002 .

Lin, Z., Zhu, W., Gu, J., Kil, J., Tensmeyer, C., Zhang, L., Liu, S., Zhang, R., Huang, L., Morariu, V.I., Sun, T., 2026b. MiLDEdit: reasoning-based multi-layer design document editing. arXiv preprint arXiv:2601.04589 .

Liu, H., Li, C., Wu, Q., Lee, Y.J., 2023. Visual instruction tuning, in: Advances in Neural Information Processing Systems.

Liu, J., Xu, S., Yang, Q., Wang, Y., Chen, X., Ji, Z., 2025a. FAPE-IR: frequency-aware planning and execution framework for all-in-one image restoration. arXiv preprint arXiv:2511.14099 .

Liu, L., Cai, C., Shen, S., Liang, J., Ouyang, W., Ye, T., Mao, J., Duan, H., Yao, J., Zhang, X., Hu, Q., Zhai, G., 2025b. MoA-VR: a mixture-of-agents system toward all-in-one video restoration. IEEE Journal of Selected Topics in Signal Processing 19, 1822–1837. doi:10.1109/JSTSP.2025.3627128.

Liu, S., Cheng, H., Liu, H., Zhang, H., Li, F., Ren, T., Zou, X., Yang, J., Su, H., Zhu, J., Zhang, L., Gao, J., et al., 2024a. LLaVA-Plus: learning to use tools for creating multimodal agents, in: European Conference on Computer Vision.

Liu, S., Han, Y., Xing, P., Yin, F., Wang, R., Cheng, W., Liao, J., Wang, Y., Fu, H., Han, C., Li, G., Peng, Y., et al., 2025c. Step1X-Edit: a practical framework for general image editing. arXiv preprint arXiv:2504.17761 .

Liu, S., Zeng, Z., Ren, T., Li, F., Zhang, H., Yang, J., Jiang, Q., Li, C., Yang, J., Su, H., Zhu, J., Zhang, L., 2024b. Grounding DINO: marrying DINO with grounded pre-training for open-set object detection, in: European Conference on Computer Vision.

Lu, J., Wu, Y., Zhao, Z., Wang, H., Jimenez, F., Majeedi, A., Fu, Y., 2026. Restore-R1: efficient image restoration agents via reinforcement learning with multimodal LLM perceptual feedback, in: 2026 IEEE/CVF Conference on Computer Vision and Pattern Recognition- FINDINGS Track (CVPRF), IEEE. pp. 8630–8640. doi:10.1109/CVPRF71909.2026.00828.

Ma, S., Guo, Y., Su, J., Huang, Q., Zhou, Z., Wang, Y., 2026. Talk2Image: a multi-agent system for multi-turn image generation and editing. Proceedings of the AAAI Conference on Artificial Intelligence 40, 32437–32445. doi:10.1609/aaai.v40i38.40519.

Ma, Y., Ji, J., Ye, K., Lin, W., Wang, Z., Zheng, Y., Zhou, Q., Sun, X., Ji, R., 2024. I2EBench: a comprehensive benchmark for instruction-based image editing, in: Advances in Neural Information Processing Systems.

Madaan, A., Tandon, N., Gupta, P., Hallinan, S., Gao, L., Wiegreffe, S., Alon, U., Dziri, N., Prabhumoye, S., Yang, Y., Gupta, S., Majumder, B.P., et al., 2023. Self-refine: Iterative refinement with self-feedback, in: Advances in Neural Information Processing Systems.

Mahdi, M., 2024. Intelligent agent-based architecture for low-light image enhancement using an A3C framework. AlKadhim Journal for Computer Science 2, 35–47. doi:10.61710/kjcs.v2i4.95.

Maier, A., Kachelriess, L., Bayer, S., Huang, Y., Xia, Y., Simpson, A., Zaiss, M., 2026. Agentic autoresearch for CT reconstruction. arXiv preprint arXiv:2607.22824 .

Mao, Q., Hu, H., She, Y., Gao, D., Chen, H., Jin, L., 2026. EmoAgent: a multi-agent framework for diverse affective image manipulation. IEEE Transactions on Affective Computing 17, 1618–1635. doi:10.1109/TAFFC.2026.3672501.

Mei, H., He, H., Qiu, H., Yang, L., Duan, L., Zhao, S., Zhan, P., Chen, Q.G., et al., 2026. Beyond visual generation: Frontiers, challenges, and future directions in agentic visual creation. Preprints.org preprint. doi:10.20944/preprints202609.0682.v1.

Mirzaei, I., 2026. Sample more, reflect less: Self-refine and reflexion lose to repeated sampling at equal token cost, from 1.5B to 7B. arXiv preprint arXiv:2607.28576 .

Mnih, V., Badia, A.P., Mirza, M., Graves, A., Lillicrap, T.P., Harley, T., Silver, D., Kavukcuoglu, K., 2016. Asynchronous methods for deep reinforcement learning, in: International Conference on Machine Learning.

Mukherjee, S., Petrangeli, S., Kveton, B., Bui, T., Dernoncourt, F., Mukherjee, A., 2026. Agentic planning with reasoning for image styling via offline RL. arXiv preprint arXiv:2603.07148 .

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., et al., 2022. Training language models to follow instructions with human feedback, in: Advances in Neural Information Processing Systems 35, Neural Information Processing Systems Foundation, Inc. (NeurIPS). pp. 27730–27744. doi:10.52202/068431-2011.

Pan, J., He, H., Bowman, S.R., Feng, S., 2024. Spontaneous reward hacking in iterative self-refinement. arXiv preprint arXiv:2407.04549 .

Podell, D., English, Z., Lacey, K., Blattmann, A., Dockhorn, T., Müller, J., Penna, J., Rombach, R., 2024. SDXL: improving latent diffusion models for high-resolution image synthesis, in: International Conference on Learning Representations.

Potlapalli, V., Zamir, S.W., Khan, S., Khan, F.S., 2023. PromptIR: prompting for all-in-one blind image restoration, in: Advances in Neural Information Processing Systems.

Pu, Y., Zheng, H., Mo, Z., Pang, Z., Zhang, H., Fan, T., Wu, S., Wei, J., 2026. CAMEO: a conditional and quality-aware multi-agent image editing orchestrator. arXiv preprint arXiv:2604.03156 .

Qi, H., Zhou, Z., Yi, J., Liu, K., Lihe, Z., Wang, J., Yoshie, O., Yokoya, N., 2026. EORestore-Agent: fidelity-guided agentic restoration of remote sensing images with composite degradations. arXiv preprint arXiv:2610.06196 .

Qin, C., Zhang, Z., Chang, Z., Guo, C., Wang, R., Hu, T., Cheng, M.M., Li, C., 2026. Dotting the eye: An intent-driven image retouching agent for visual focus enhancement, in: Computer Vision – ECCV 2026, Springer Nature Switzerland. pp. 574–591. doi:10.1007/978-3-032-37490-5_32.

Qiu, J., Lu, P., Peng, X., Guo, W., Zhao, Z., Dong, X., 2024. Learning realistic sketching: A dual-agent reinforcement learning approach, in: Proceedings of the 32nd ACM International Conference on Multimedia, ACM. pp. 5921–5929. doi:10.1145/3664647.3680759.

Qiu, Z., Chen, K., Wang, X., Xia, Y., Seneviratne, S., Halgamuge, S., 2026. MSRAMIE: multimodal structured reasoning agent for multi-instruction image editing. arXiv preprint arXiv:2603.16967 .

Rafailov, R., Sharma, A., Mitchell, E., Manning, C.D., Ermon, S., Finn, C., 2023. Direct preference optimization: Your language model is secretly a reward model, in: Advances in Neural Information Processing Systems 36, Neural Information Processing Systems Foundation, Inc. (NeurIPS). pp. 53728–53741. doi:10.52202/075280-2338.

Rajan, A.S., Singh, K.K., Lee, Y.J., 2026. From plans to pixels: Learning to plan and orchestrate for open-ended image editing. arXiv preprint arXiv:2605.15181 .

Robitza, W., 2026. JEVQA: Video quality from metadata, bitstream, and pixel features with a general-purpose decision model. arXiv preprint arXiv:2609.24395 .

Rombach, R., Blattmann, A., Lorenz, D., Esser, P., Ommer, B., 2022. High-resolution image synthesis with latent diffusion models, in: 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE. pp. 10674–10685. doi:10.1109/CVPR52688.2022.01042.

Russell, S., Norvig, P., 2021. Artificial Intelligence: A Modern Approach. 4 ed., Pearson.

Saharia, C., Ho, J., Chan, W., Salimans, T., Fleet, D.J., Norouzi, M., 2022. Image super-resolution via iterative refinement. IEEE Transactions on Pattern Analysis and Machine Intelligence , 1–14doi:10.1109/TPAMI.2022.3204461.

Schick, T., Dwivedi-Yu, J., Dessi, R., Raileanu, R., Lomeli, M., Zettlemoyer, L., Cancedda, N., Scialom, T., 2023. Toolformer: Language models can teach themselves to use tools, in: Advances in Neural Information Processing Systems.

Schulman, J., Wolski, F., Dhariwal, P., Radford, A., Klimov, O., 2017. Proximal policy optimization algorithms. arXiv preprint arXiv:1707.06347 .

Shao, J., Hu, S., Zhang, X., Song, B., Jing, Y., Wu, X., Wan, J., 2026. SkillIR: evolving scene-aware skills for agentic image restoration. arXiv preprint arXiv:2609.21468 .

Shao, Z., Wang, P., Zhu, Q., Xu, R., Song, J., Bi, X., Zhang, H., Zhang, M., Li, Y.K., Wu, Y., Guo, D., 2024. DeepSeekMath: pushing the limits of mathematical reasoning in open language models. arXiv preprint arXiv:2402.03300 .

Shen, F., Xie, C., Wang, L., Zhang, Z., Jiang, X., Du, X., Tang, J., 2026a. IMAGAgent: orchestrating multi-turn image editing via constraint-aware planning and reflection. arXiv preprint arXiv:2603.29602 .

Shen, S., Liang, J., Cai, C., Geng, C., Duan, H., Zhang, X., Hu, Q., Zhai, G., 2026b. Agentic retoucher for text-to-image generation, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition.

Shen, Y., Song, K., Tan, X., Li, D., Lu, W., Zhuang, Y., 2023. HuggingGPT: solving AI tasks with ChatGPT and its friends in hugging face, in: Advances in Neural Information Processing Systems.

Sheynin, S., Polyak, A., Singer, U., Kirstain, Y., Zohar, A., Ashual, O., Parikh, D., Taigman, Y., 2024. Emu edit: Precise image editing via recognition and generation tasks, in: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE. pp. 8871–8879. doi:10.1109/CVPR52733.2024.00847.

Shinn, N., Cassano, F., Gopinath, A., Narasimhan, K., Yao, S., 2023. Reflexion: language agents with verbal reinforcement learning, in: Advances in Neural Information Processing Systems 36, Neural Information Processing Systems Foundation, Inc. (NeurIPS). pp. 8634–8652. doi:10.52202/075280-0377.

Shuai, X., Ding, H., Ma, X., Tu, R., Jiang, Y.G., Tao, D., 2024. A survey of multimodal-guided image editing with text-to-image diffusion models. arXiv preprint arXiv:2406.14555 .

Shukla, M., 2026. Adaptive monitoring and real-world evaluation of agentic AI systems. AI and Ethics 6. doi:10.1007/s43681-026-00990-y.

Silver, D., Huang, A., Maddison, C.J., Guez, A., Sifre, L., van den Driessche, G., Schrittwieser, J., Antonoglou, I., Panneershelvam, V., Lanctot, M., Dieleman, S., Grewe, D., et al., 2016. Mastering the game of go with deep neural networks and tree search. Nature 529, 484–489. doi:10.1038/nature16961.

Song, X., Shangguan, J., Li, Y., Zhang, J., Ren, J., Bai, R., Chen, X., Jiang, X., 2025. CEARI: co-evolutionary agents for reassembling and inpainting puzzles with gaps and missing pieces, in: Proceedings of the 33rd ACM International Conference on Multimedia, ACM. pp. 2634–2642. doi:10.1145/3746027.3754695.

Su, J., Lan, Q., Wang, Z., Xia, Y., Wen, H., Duan, Y., Xiao, X., Shi, T., Jingsong, Y., He, L., 2026. ComfySearch: autonomous exploration and reasoning for ComfyUI workflows. arXiv preprint arXiv:2601.04060 .

Sumers, T.R., Yao, S., Narasimhan, K., Griffiths, T.L., 2024. Cognitive architectures for language agents. Transactions on Machine Learning Research .

Sun, S., Chen, R., Liu, Z., Liu, X., Qian, C., Ren, W., Cao, X., 2026a. CaptchaAgent: verifiable visual reasoning for CAPTCHA solving. arXiv preprint arXiv:2610.12866 .

Sun, W., Yin, X., Xiao, H., Gu, J., Dong, C., 2026b. From specialized models to agentic systems: Progress and challenges of agents in image restoration. Journal of Integration Technology 15, 1–21. doi:10.12146/j.issn.2095-3135.20251115001. in Chinese.

Surís, D., Menon, S., Vondrick, C., 2023. ViperGPT: visual inference via python execution for reasoning, in: 2023 IEEE/CVF International Conference on Computer Vision (ICCV), IEEE. pp. 11854–11864. doi:10.1109/ICCV51070.2023.01092.

Sutton, R.S., Barto, A.G., 2018. Reinforcement Learning: An Introduction. 2 ed., MIT Press.

Tang, K., Arunshankar, P., Hua, A., Yang, A., Qin, Y., 2026. Banana100: breaking NR-IQA metrics by 100 iterative image replications with nano banana pro. arXiv preprint arXiv:2604.03400 .

Tianyidan, X., Ma, R., Wang, Q., Ye, X., Liu, F., Tai, Y., Zhang, Z., Wang, L., Yi, Z., 2025. Anywhere: A multi-agent framework for user-guided, reliable, and diverse foreground-conditioned image generation. Proceedings of the AAAI Conference on Artificial Intelligence 39, 7410–7418. doi:10.1609/aaai.v39i7.32797.

Ulhaq, A., 2025. Agentic AI for computer vision: A review. engrXiv preprint. doi:10.31224/5832.

Venkatesh, K., Dunlop, C., Yanardag, P., 2025. CREA: a collaborative multi-agent framework for creative image editing and generation, in: Advances in Neural Information Processing Systems 38, Neural Information Processing Systems Foundation, Inc. (NeurIPS). pp. 189799–189859. doi:10.52202/085713-5705.

Vinker, Y., Shaham, T.R., Zheng, K., Zhao, A., Fan, J.E., Torralba, A., 2025. SketchAgent: language-driven sequential sketch generation, in: 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE. pp. 23355–23368. doi:10.1109/CVPR52734.2025.02175.

Wan, X., Zhou, H., Sun, R., Nakhost, H., Jiang, K., Sinha, R., Arık, S.Ö., 2025. Maestro: Self-improving text-to-image generation via agent orchestration. arXiv preprint arXiv:2509.10704 .

Wang, G., Xie, Y., Jiang, Y., Mandlekar, A., Xiao, C., Zhu, Y., Fan, L., Anandkumar, A., 2024a. Voyager: An open-ended embodied agent with large language models. Transactions on Machine Learning Research .

Wang, J., Chan, K.C.K., Loy, C.C., 2023a. Exploring CLIP for assessing the look and feel of images, in: AAAI Conference on Artificial Intelligence.

Wang, K., Chen, R., Zheng, T., Huang, H., 2025a. ImAgent: a unified multimodal agent framework for test-time scalable image generation. arXiv preprint arXiv:2511.11483 .

Wang, L., Ma, C., Feng, X., Zhang, Z., Yang, H., Zhang, J., Chen, Z., Tang, J., Chen, X., Lin, Y., Zhao, W.X., Wei, Z., et al., 2024b. A survey on large language model based autonomous agents. Frontiers of Computer Science 18. doi:10.1007/s11704-024-40231-1.

Wang, P., Zhang, Z., Gao, G., Zhang, Y., Zheng, Z., 2025b. AgentPolyp: accurate polyp segmentation via image enhancement agent. IEEE Signal Processing Letters 32, 3062–3066. doi:10.1109/LSP.2025.3591721.

Wang, X., Li, Y., Zhang, H., Shan, Y., 2021a. Towards real-world blind face restoration with generative facial prior, in: 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE. pp. 9164–9174. doi:10.1109/CVPR46437.2021.00905.

Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., Chowdhery, A., Zhou, D., 2023b. Self-consistency improves chain of thought reasoning in language models, in: International Conference on Learning Representations.

Wang, X., Xie, L., Dong, C., Shan, Y., 2021b. Real-ESRGAN: training real-world blind super-resolution with pure synthetic data, in: 2021 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW), IEEE. pp. 1905–1914. doi:10.1109/ICCVW54120.2021.00217.

Wang, Y., Wen, R., Ishii, H., Ohya, J., 2025c. Multi-weather DomainShifter: a comprehensive multi-weather transfer LLM agent for handling domain shift in aerial image processing. Journal of Imaging 11, 395. doi:10.3390/jimaging11110395.

Wang, Y., Yan, Q., Zhou, J., Dai, D., Dong, W., 2026. PaAgent: portrait-aware image restoration agent via subjective-objective reinforcement learning. arXiv preprint arXiv:2603.17055 .

Wang, Z., Li, A., Li, Z., Liu, X., 2024c. GenArtist: multimodal LLM as an agent for unified image generation and editing, in: Advances in Neural Information Processing Systems.

Wang, Z., Xie, E., Li, A., Wang, Z., Liu, X., Li, Z., 2024d. Divide and conquer: Language models can plan and self-correct for compositional text-to-image generation. arXiv preprint arXiv:2401.15688 .

Wei, J., Tan, C., Chen, Q., Wu, G., Li, S., Gao, Z., Sun, L., Yu, B., Guo, R., 2025. From words to structured visuals: A benchmark and framework for text-to-diagram generation and editing, in: 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE. pp. 13315–13325. doi:10.1109/CVPR52734.2025.01243.

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q.V., Zhou, D., 2022. Chain-of-thought prompting elicits reasoning in large language models, in: Advances in Neural Information Processing Systems 35, Neural Information Processing Systems Foundation, Inc. (NeurIPS). pp. 24824–24837. doi:10.52202/068431-1800.

Wei, Y., Zhang, Y., Zheng, H., Ren, J., Xu, X., Shi, Z., Zhang, Z., Wang, M., 2026. IAMAgent: toward an interactive and adaptive multi-agent system for image restoration. IEEE Transactions on Image Processing 35, 3127–3140. doi:10.1109/TIP.2026.3671594.

Wei, Y., Zhang, Z., Ren, J., Xu, X., Hong, R., Yang, Y., Yan, S., Wang, M., 2023. Clarity ChatGPT: an interactive and adaptive processing system for image restoration and enhancement. arXiv preprint arXiv:2311.11695 .

Wooldridge, M., Jennings, N.R., 1995. Intelligent agents: theory and practice. The Knowledge Engineering Review 10, 115–152. doi:10.1017/S0269888900008122.

Wu, C., Li, J., Zhou, J., Lin, J., Gao, K., Yan, K., Yin, S.m., Bai, S., Xu, X., Chen, Y., Chen, Y., Tang, Z., et al., 2025. Qwen-image technical report. arXiv preprint arXiv:2508.02324 .

Wu, C., Yin, S., Qi, W., Wang, X., Tang, Z., Duan, N., 2023a. Visual ChatGPT: talking, drawing and editing with visual foundation models. arXiv preprint arXiv:2303.04671 .

Wu, H., Zhang, Z., Zhang, E., Chen, C., Liao, L., Wang, A., Li, C., Sun, W., Yan, Q., Zhai, G., Lin, W., 2024a. Q-bench: A benchmark for general-purpose foundation models on low-level vision, in: International Conference on Learning Representations.

Wu, H., Zhang, Z., Zhang, E., Chen, C., Liao, L., Wang, A., Xu, K., Li, C., Hou, J., Zhai, G., Xue, G., Sun, W., et al., 2024b. Q-instruct: Improving low-level visual abilities for multi-modality foundation models, in: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE. pp. 25490–25500. doi:10.1109/CVPR52733.2024.02408.

Wu, H., Zhang, Z., Zhang, W., Chen, C., Liao, L., Li, C., Gao, Y., Wang, A., Zhang, E., Sun, W., Yan, Q., Min, X., et al., 2024c. Q-align: Teaching LMMs for visual scoring via discrete text-defined levels, in: International Conference on Machine Learning.

Wu, Q., Shi, J., Jenni, S., Kafle, K., Wang, T., Chang, S., Zhao, H., 2026. RetouchIQ: MLLM agents for instruction-based image retouching with generalist reward, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition.

Wu, T.H., Lian, L., Gonzalez, J.E., Li, B., Darrell, T., 2024d. Self-correcting LLM-Controlled diffusion models, in: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE. pp. 6327–6336. doi:10.1109/CVPR52733.2024.00605.

Wu, X., Hao, Y., Sun, K., Chen, Y., Zhu, F., Zhao, R., Li, H., 2023b. Human preference score v2: a solid benchmark for evaluating human preferences of text-to-image synthesis. arXiv preprint arXiv:2306.09341 .

Xi, Z., Chen, W., Guo, X., He, W., Ding, Y., Hong, B., Zhang, M., Wang, J., Jin, S., Zhou, E., Zheng, R., Fan, X., et al., 2025. The rise and potential of large language model based agents: a survey. Science China Information Sciences 68. doi:10.1007/s11432-024-4222-0.

Xie, J., Chen, Z., Zhang, R., Li, G., 2025. Large multimodal agents: a survey. Visual Intelligence 3. doi:10.1007/s44267-025-00093-y.

Xing, Z., Wang, J., Xia, R., Ye, K., Qu, Z., Yi, B., Liu, T., Chen, J., et al., 2026. Agentic visual generation: A survey. Preprints.org preprint. doi:10.20944/preprints202609.0173.v1.

Xu, J., Liu, X., Wu, Y., Tong, Y., Li, Q., Ding, M., Tang, J., Dong, Y., 2023. ImageReward: learning and evaluating human preferences for text-to-image generation, in: Advances in Neural Information Processing Systems.

Xu, W., Zhu, G., Zhao, X., Pan, L., Li, L., Wang, W., 2024. Pride and prejudice: LLM amplifies self-bias in self-refinement, in: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics. pp. 15474–15492. doi:10.18653/v1/2024.acl-long.826.

Xu, Z., Duan, H., Nie, Y., Du, M., Wu, S., Min, X., Zheng, T., Zhang, J., Xu, S., Chen, J., Li, B., Zhai, G., 2026a. EditRefiner: a human-aligned agentic framework for image editing refinement. arXiv preprint arXiv:2605.07457 .

Xu, Z., Wang, Y., Yang, X., Wang, L., Luo, W., Zhang, K., Hu, B., Zhang, M., 2026b. ComfyUI-R1: exploring reasoning models for workflow generation, in: Findings of the Association for Computational Linguistics: ACL 2026, pp. 2999–3013.

Xu, Z., Yang, X., Wang, Y., Hu, Q., Wu, Z., Wang, L., Luo, W., Zhang, K., Hu, B., Zhang, M., 2025. ComfyUI-Copilot: an intelligent assistant for automated workflow development, in: Annual Meeting of the Association for Computational Linguistics: System Demonstrations.

Xue, X., Lu, Z., Huang, D., Wang, Z., Ouyang, W., Bai, L., 2025. ComfyBench: benchmarking LLM-based agents in ComfyUI for autonomously designing collaborative AI systems, in: 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE. pp. 24614–24624. doi:10.1109/CVPR52734.2025.02292.

Yan, W., Zhang, R., Tang, K., Cheng, Y., Liu, Q., 2026. RIRF: reasoning image restoration framework. arXiv preprint arXiv:2604.09511 .

Yang, L., Yu, Z., Meng, C., Xu, M., Ermon, S., Cui, B., 2024a. Mastering text-to-image diffusion: Recaptioning, planning, and generating with multimodal LLMs, in: International Conference on Machine Learning.

Yang, S., Wu, T., Shi, S., Lao, S., Gong, Y., Cao, M., Wang, J., Yang, Y., 2022. MANIQA: multi-dimension attention network for no-reference image quality assessment, in: 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), IEEE. pp. 1190–1199. doi:10.1109/CVPRW56347.2022.00126.

Yang, Y., Li, C., Chai, Z., Dong, X., Jin, Z., 2026. Agentic image restoration links closed-loop controllers with protocol-aware evaluation. Preprints.org preprint. doi:10.20944/preprints202609.1256.v1.

Yang, Z., Li, L., Wang, J., Lin, K., Azarnasab, E., Ahmed, F., Liu, Z., Liu, C., Zeng, M., Wang, L., 2023. MM-REACT: prompting ChatGPT for multimodal reasoning and action. arXiv preprint arXiv:2303.11381 .

Yang, Z., Wang, J., Li, L., Lin, K., Lin, C.C., Liu, Z., Wang, L., 2024b. Idea2Img: iterative self-refinement with GPT-4V(ision) for automatic image design and generation, in: European Conference on Computer Vision.

Yao, M., You, Z., Tam, K.M., Wang, M., Xue, T., 2026. PhotoAgent: exploratory visual aesthetic planning with large vision models, in: International Conference on Machine Learning.

Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T.L., Cao, Y., Narasimhan, K., 2023a. Tree of thoughts: Deliberate problem solving with large language models, in: Advances in Neural Information Processing Systems.

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., Cao, Y., 2023b. ReAct: synergizing reasoning and acting in language models, in: International Conference on Learning Representations.

Ye, R., Zhang, J., Liu, Z., Zhu, Z., Yang, S., Li, L., Fu, T., Dernoncourt, F., Zhao, Y., Zhu, J., Rossi, R., Chai, W., et al., 2026. Agent banana: High-fidelity image editing with agentic thinking and tooling. arXiv preprint arXiv:2602.09084 .

Ye, Y., He, X., Li, Z., Lin, B., Yuan, S., Yan, Z., Hou, B., Yuan, L., 2025. ImgEdit: a unified image editing dataset and benchmark, in: Advances in Neural Information Processing Systems.

You, Z., Gu, J., Cai, X., Li, Z., Zhu, K., Dong, C., Xue, T., 2025. Enhancing descriptive image quality assessment with a large-scale multi-modal dataset. IEEE Transactions on Image Processing 34, 8201–8215. doi:10.1109/TIP.2025.3639998.

You, Z., Li, Z., Gu, J., Yin, Z., Xue, T., Dong, C., 2024. Depicting beyond scores: Advancing image quality assessment through multi-modal language models, in: European Conference on Computer Vision.

Yu, B., Ozdemir, S., Xing, Y., Hashimoto, F., Wu, J., Chen, Y., Rominger, A., Fang, R., Shi, K., Pan, T., Gong, K., 2026a. VLM- and LLM-Driven multi-agent system for PET image denoising. arXiv preprint arXiv:2608.13791 .

Yu, F., Gu, J., Li, Z., Hu, J., Kong, X., Wang, X., He, J., Qiao, Y., Dong, C., 2024. Scaling up to excellence: Practicing model scaling for photo-realistic image restoration in the wild, in: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE. pp. 25669–25680. doi:10.1109/CVPR52733.2024.02425.

Yu, G., Yao, Y., 2026. Visual Jev: accurate and efficient decisions from shared visual context. arXiv preprint arXiv:2609.25845 .

Yu, J., Xiao, J., Zhu, C., Li, J., Li, J., Deng, H., Wang, X., Jia, G., Li, J., Bai, X., Zhou, B., Ma, Z., 2026b. I2E: from image pixels to actionable interactive environments for text-guided image editing, in: Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics. pp. 44827–44850. doi:10.18653/v1/2026.acl-long.2076.

Yu, K., Dong, C., Lin, L., Loy, C.C., 2018. Crafting a toolchain for image restoration by deep reinforcement learning, in: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, IEEE. pp. 2443–2452. doi:10.1109/CVPR.2018.00259.

Yu, Z., Chen, X., Li, R., Geng, Z., Sun, G., Li, H., Jiang, K., 2026c. Derain-agent: A plug-and-play agent framework for rainy image restoration. arXiv preprint arXiv:2603.11866 .

Yun, J., Park, J., Lim, H., Hyung, J., Chung, H., Choo, J., 2026. ReDesign: recovering editable design structures from images via agentic decomposition, in: Computer Vision – ECCV 2026, Springer Nature Switzerland. pp. 288–304. doi:10.1007/978-3-032-37255-0_16.

Zamir, S.W., Arora, A., Khan, S., Hayat, M., Khan, F.S., Yang, M.H., 2022. Restormer: Efficient transformer for high-resolution image restoration, in: 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE. pp. 5718–5729. doi:10.1109/CVPR52688.2022.00564.

Zeng, Z., Hua, H., Luo, J., 2026. MIRA: multimodal iterative reasoning agent for image editing, in: 2026 IEEE/CVF Conference on Computer Vision and Pattern Recognition- FINDINGS Track (CVPRF), IEEE. pp. 9564–9574. doi:10.1109/CVPRF71909.2026.00916.

Zhang, H., Liu, Z., Niu, L., Liu, J., Li, H., Cao, Z., Chen, W., Zhao, C., Meng, F., 2026. WeAgent-MMGenEdit: a full-stack recipe for multimodal agentic image generation and editing. arXiv preprint arXiv:2609.05171 .

Zhang, K., Mo, L., Chen, W., Sun, H., Su, Y., 2023a. MagicBrush: a manually annotated dataset for instruction-guided image editing, in: Advances in Neural Information Processing Systems.

Zhang, K., Zuo, W., Chen, Y., Meng, D., Zhang, L., 2017. Beyond a gaussian denoiser: Residual learning of deep CNN for image denoising. IEEE Transactions on Image Processing 26, 3142–3155. doi:10.1109/TIP.2017.2662206.

Zhang, L., Rao, A., Agrawala, M., 2023b. Adding conditional control to text-to-image diffusion models, in: 2023 IEEE/CVF International Conference on Computer Vision (ICCV), IEEE. pp. 3813–3824. doi:10.1109/ICCV51070.2023.00355.

Zhang, R., Isola, P., Efros, A.A., Shechtman, E., Wang, O., 2018. The unreasonable effectiveness of deep features as a perceptual metric, in: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, IEEE. pp. 586–595. doi:10.1109/CVPR.2018.00068.

Zhang, W., Li, X., Chen, X., Qiao, Y., Wu, X.M., Dong, C., 2024. SEAL: a framework for systematic evaluation of real-world super-resolution, in: International Conference on Learning Representations.

Zhao, B., Guo, K., Du, R., Sun, H., Wang, P., Yang, H., Gai, K., Cao, Y., Ji, W., 2026a. Making image editing easier via adaptive task reformulation with agentic executions. arXiv preprint arXiv:2604.15917 .

Zhao, L., Yang, Y., Zhang, K., Shao, W., Zhang, Y., Qiao, Y., Luo, P., Ji, R., 2024. DiffAgent: fast and accurate text-to-image API selection with large language model, in: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE. pp. 6390–6399. doi:10.1109/CVPR52733.2024.00611.

Zhao, X., Chen, X., Liu, B., Liu, Z., Zhao, Z., Gu, X., 2026b. Value-aligned prompt moderation via zero-shot agentic rewriting for safe image generation, in: AAAI Conference on Artificial Intelligence.

Zhao, Y., Ye, Y., Liu, X., Shieh, M.Q., Bui, T., 2026c. ImageEdit-R1: boosting multi-agent image editing via reinforcement learning. arXiv preprint arXiv:2603.08059 .

Zheng, L., Chiang, W.L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E., Zhang, H., Gonzalez, J., et al., 2023. Judging LLM-as-a-Judge with MT-Bench and chatbot arena, in: Advances in Neural Information Processing Systems 36, Neural Information Processing Systems Foundation, Inc. (NeurIPS). pp. 46595–46623. doi:10.52202/075280-2020.

Zhou, Y., Cao, J., Wen, F., Zhang, Z., Zhou, Y., Shi, Y., Liu, X., Timofte, R., Gool, L.V., Zhai, G., 2025. Q-agent: Quality-driven chain-of-thought image restoration agent through robust multimodal large language model. arXiv preprint arXiv:2504.07148 .

Zhu, F., Xie, S., Zeng, Y., Liu, M., Zuo, W., 2026a. OPERA: an agent for image restoration with end-to-end joint planning-execution optimization. arXiv preprint arXiv:2605.22104 .

Zhu, H., Tian, Y., Ding, K., Chen, B., Chen, B., Wang, S., Lin, W., 2025a. AgenticIQA: an agentic framework for adaptive and interpretable image quality assessment. arXiv preprint arXiv:2509.26006 .

Zhu, K., Gu, J., You, Z., Qiao, Y., Dong, C., 2025b. An intelligent agentic system for complex image restoration problems, in: International Conference on Learning Representations.

Zhu, Z., Sun, Y., Zhu, M., Ma, W., Zhang, S., Wang, Z., Yang, Z., Zhang, D., Lan, K., Zhao, Z., Liu, D., Xiang, S., et al., 2026b. IEA: amateur-friendly conversational image editing agent via three stages of multitask alignment. arXiv preprint arXiv:2606.08016 .

Zhuang, K., Wu, J., Jin, Z., 2026. EvoIR-Agent: self-evolving image restoration agentic system via experience-driven learning. arXiv preprint arXiv:2605.22208 .

Zuo, Y., Zheng, Q., Wu, M., Jiang, X., Li, R., Wang, J., Zhang, Y., Mai, G., Wang, L.V., Zou, J., Wang, X., Yang, M.H., et al., 2025. 4KAgent: agentic any image to 4K super-resolution, in: Advances in Neural Information Processing Systems.